# Symmetry Definitions

Motif: the fundamental part of a symmetric design that, when repeated, creates the whole pattern

Operation: some act that reproduces the motif to create the pattern

Element: an operation located at a particular point in space

# What are the motif and operation?

Rotation through an angle about a certain axis

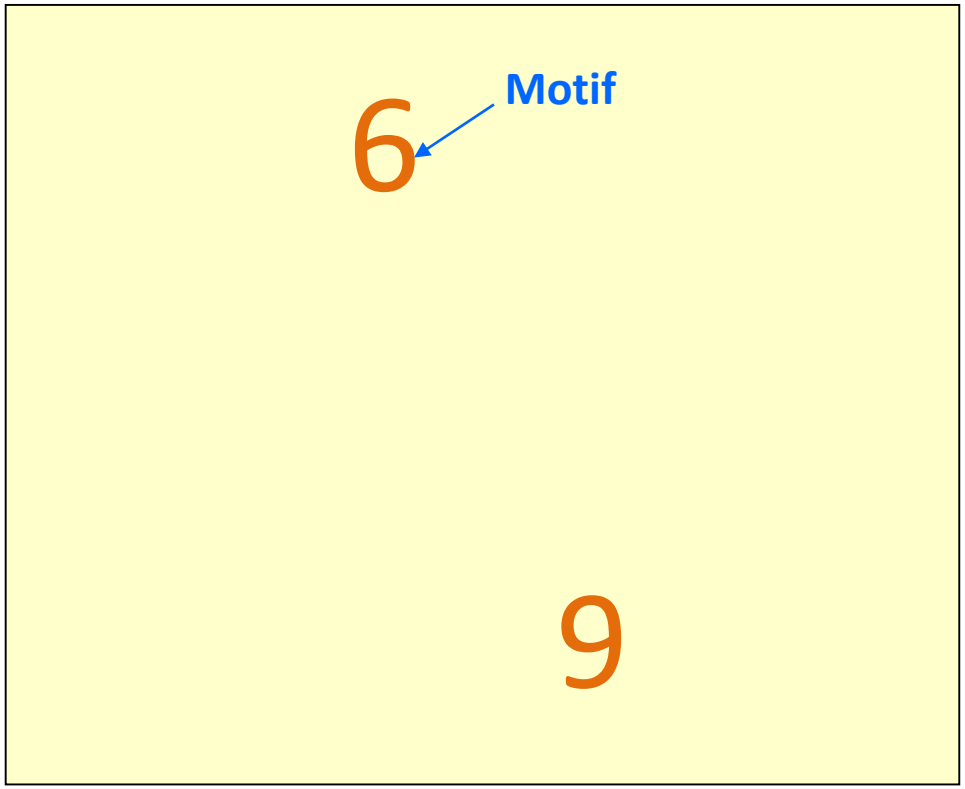
Operation

a. Two-fold rotation

= 360$^{o}$/2 rotation

to reproduce a motif in a symmetrical pattern

A Symmetrical Pattern



# Practice: 2-D Symmetry

a. Two-fold rotation

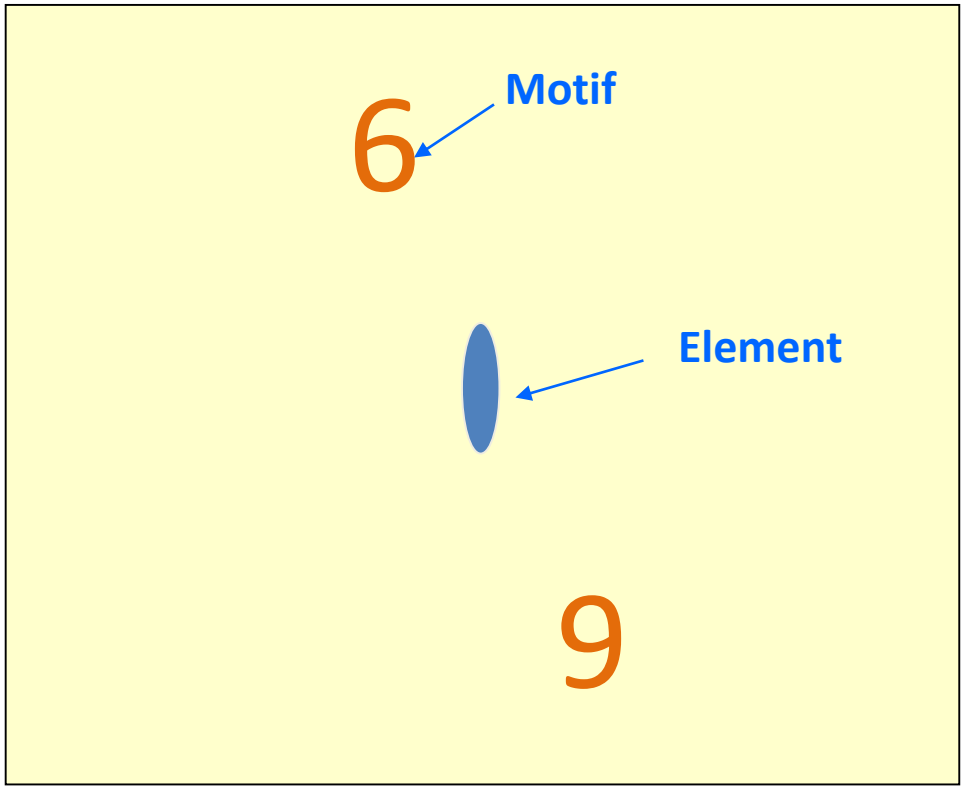
Operation

= 360°/2 rotation

to reproduce a motif in a symmetrical pattern



= the symbol for a two-fold rotation



# 2-D Symmetry

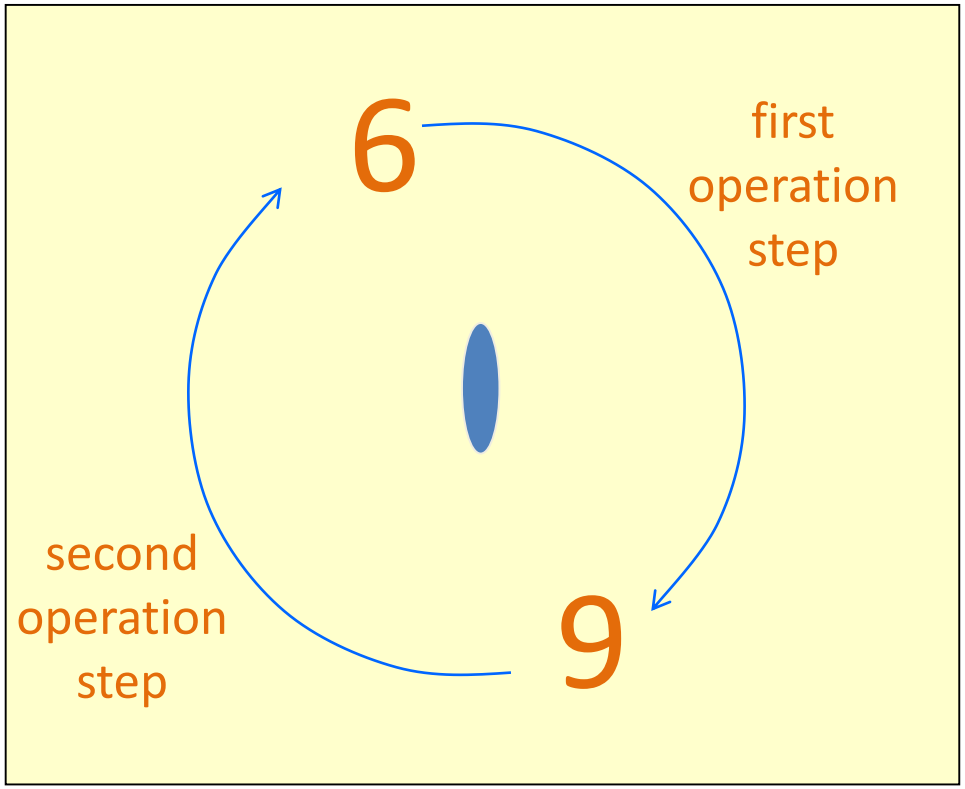
a. Two-fold rotation

= 360°/2 rotation

to reproduce a motif in a symmetrical pattern



= the symbol for a two-fold rotation

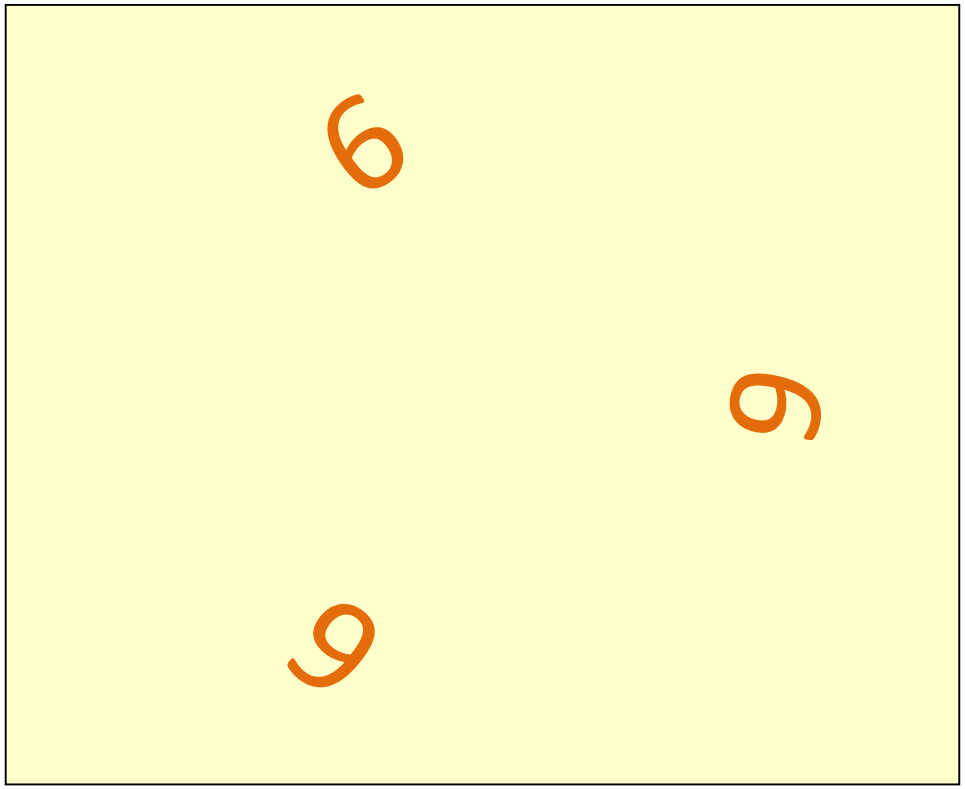


# 2-D Symmetry

b. Three-fold rotation

= $360^{o}/3$ rotation

to reproduce a motif in a symmetrical pattern

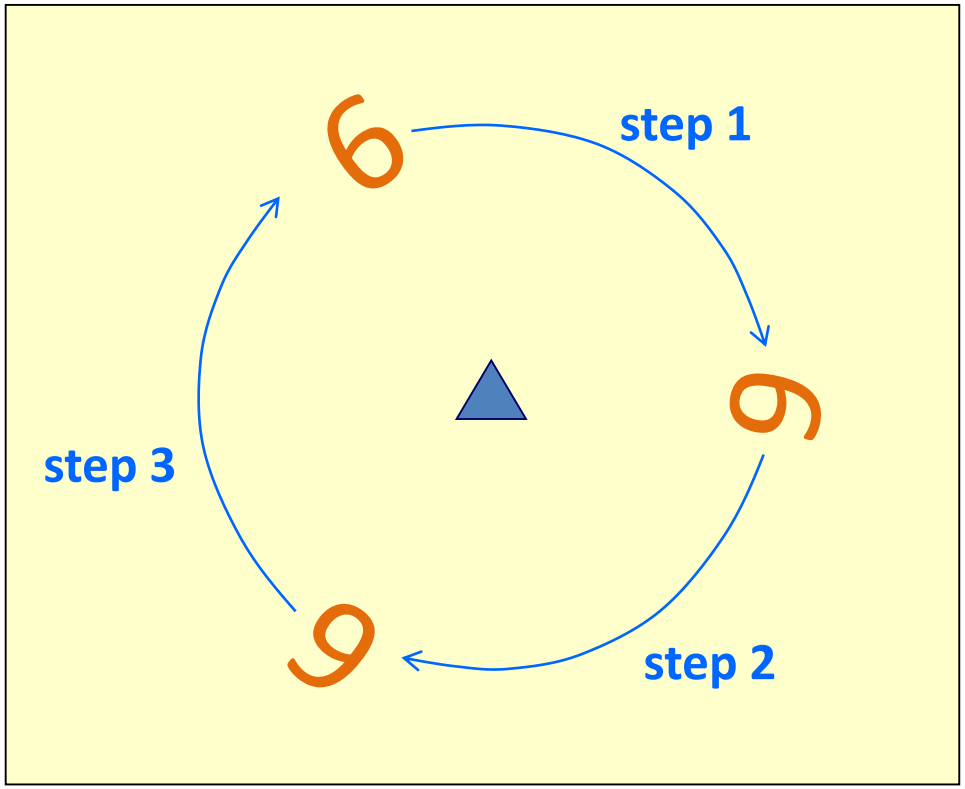


# 2-D Symmetry

b. Three-fold rotation

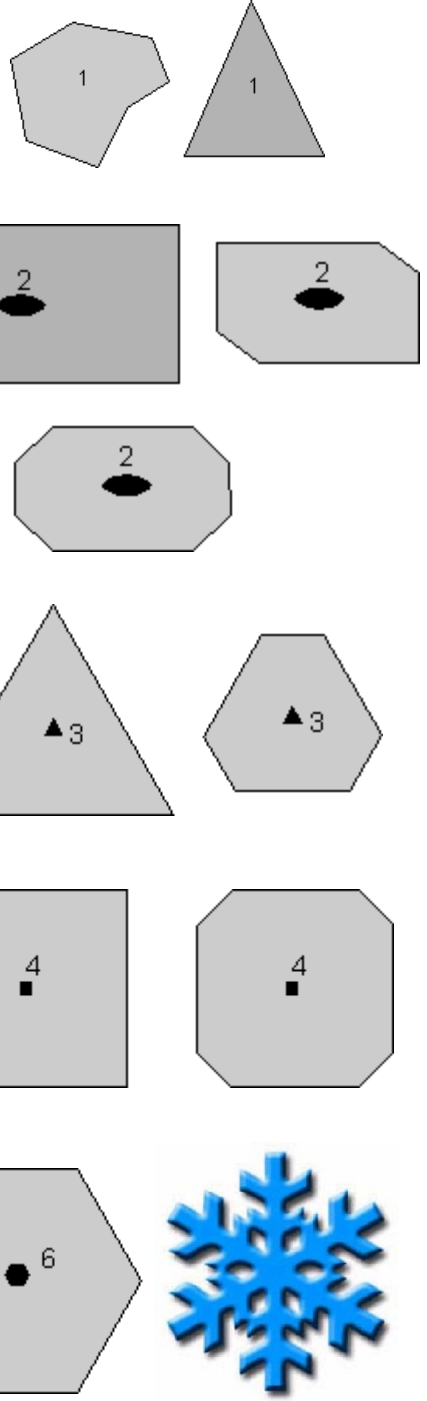
= 360°/3 rotation

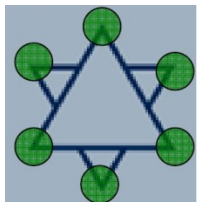
to reproduce a motif in a symmetrical pattern



# Rotation Axes

- Trivial case is 360° rotation
- Order of rotation: 2-, 3-, 4-, and 6- correspond to 180°, 120°, 90°, and 60°.
  - These are only symmetry rotations allowed in crystals with long-range order
  - Small aggregates (short-range order) or molecules can also have 5-, 7-, etc. fold rotational symmetry



Rotational Symmetry Link
http://www.staff.ncl.ac.uk/j.p.goss/symmetry/Molecules_pov.html

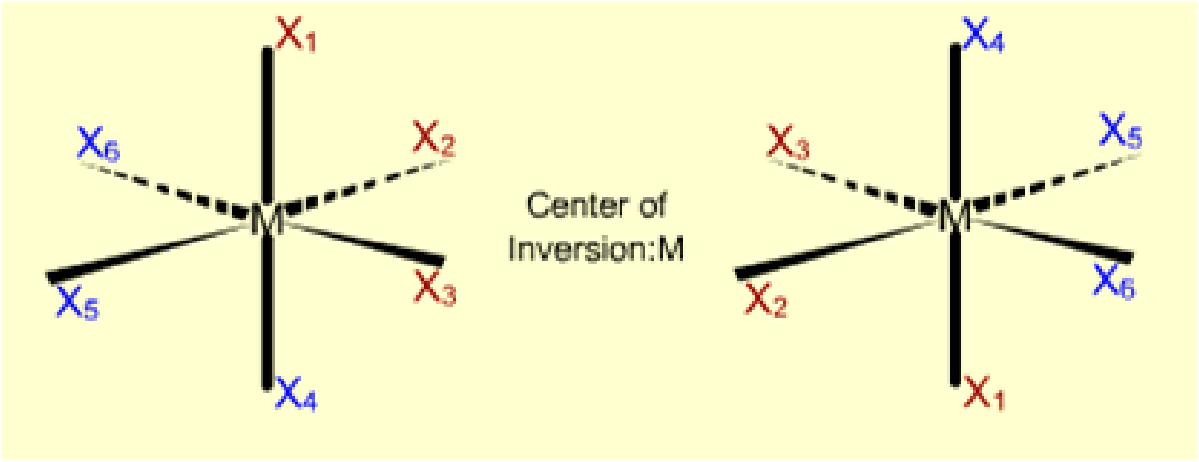
# Inversion Center

$$I = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

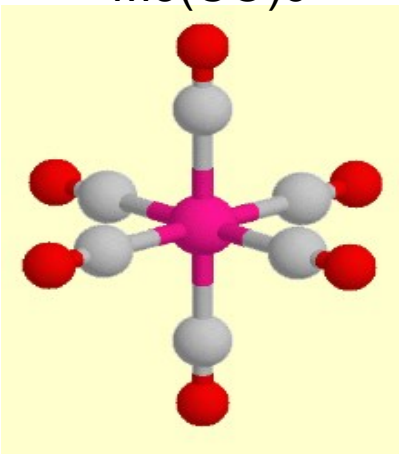
- Center of symmetry: A point at the center of a molecule.

(x,y,z) --> (-x,-y,-z)

- It is not necessary to have an atom in the center (benzene, ethane).
- Tetrahedral, triangles, pentagons do not have a center of inversion symmetry.
- All Bravais **lattices** are inversion symmetric.
- Once you add the basis, this symmetry may be lost.




Mo(CO)6



# 2-D Symmetry

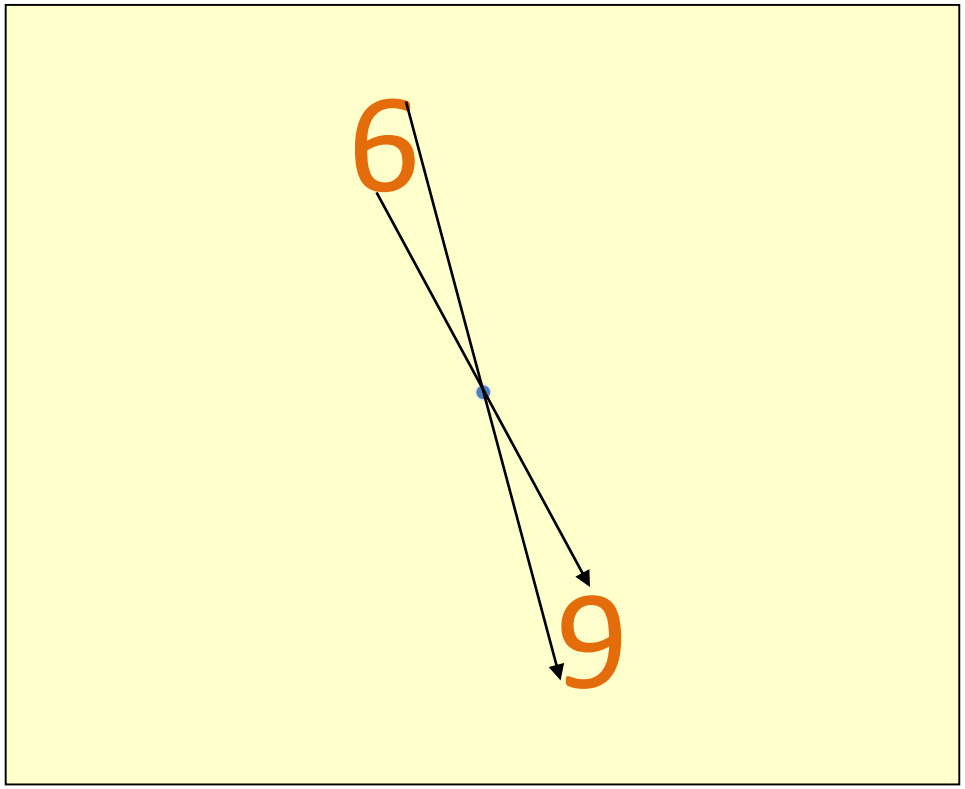
Symmetry Elements

2. Inversion (*i*)

inversion through a center to reproduce a motif in a symmetrical pattern

- = symbol for an inversion center

Inversion is identical to 2-fold rotation in 2-D, but is unique in 3-D



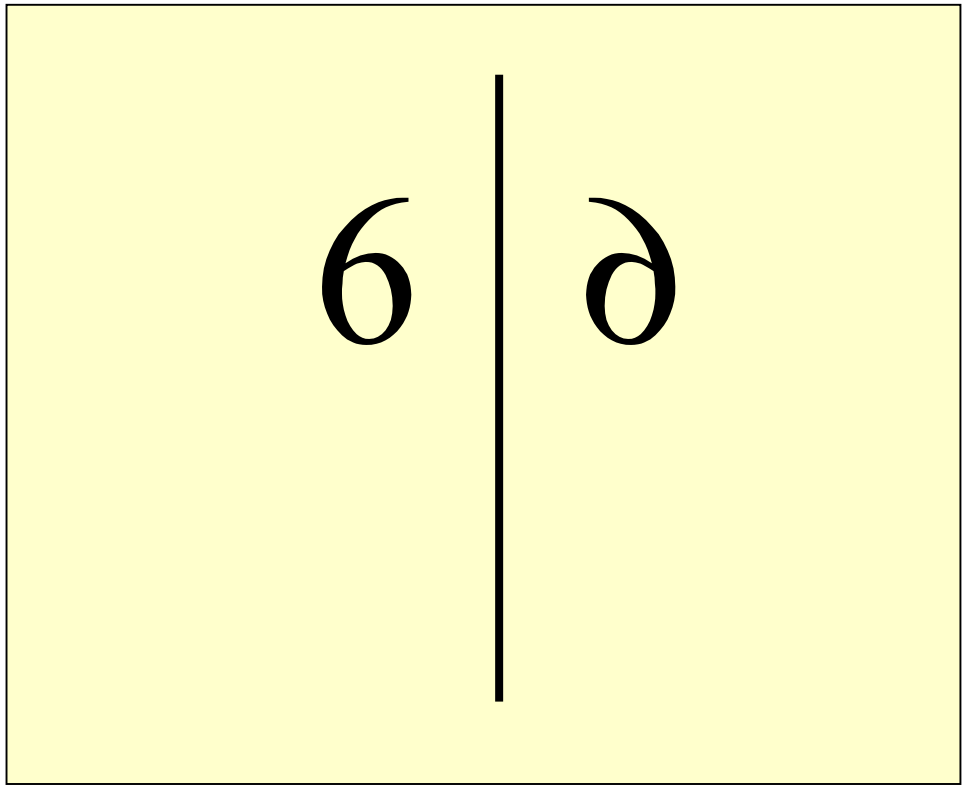
# 2-D Symmetry

Symmetry Elements

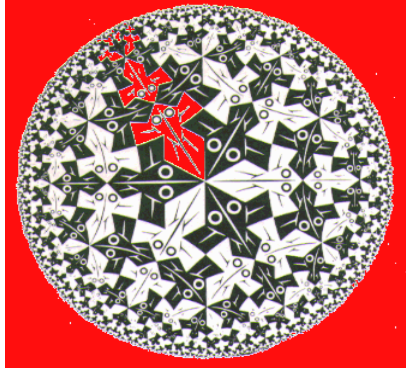
3. Reflection (m)

Reflection across a "mirror plane" reproduces a motif

= symbol for a mirror plane



(Escher, Circle limit I)

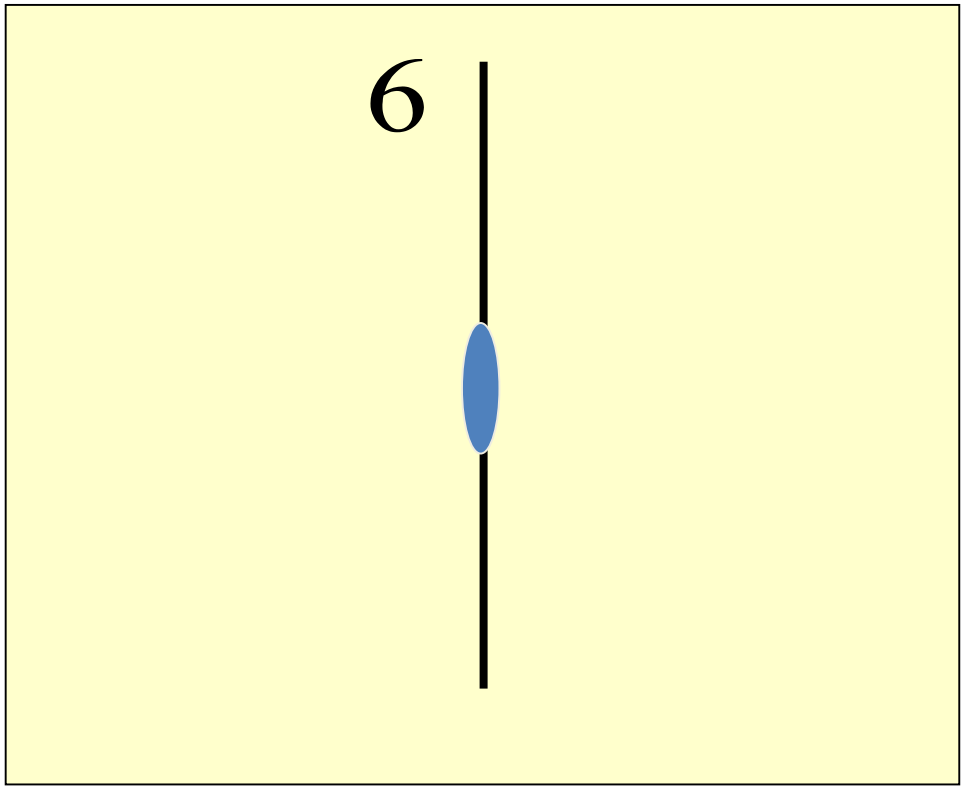


# Crystal Symmetry

- We now have 6 unique 2-D symmetry operations:
  1 2 3 4 6 m (inversion not unique)
- Combinations of symmetry elements are also possible
- The group of operations that can map a crystal into itself defines the crystal symmetry
- Groups act upon sets of objects (e.g., basis/motif)
- Groups with lots of operations have high symmetry
- In the interest of clarity and ease of illustration, let's start with 2-D examples

# 2-D Symmetry

Try combining a 2-fold rotation axis with a mirror

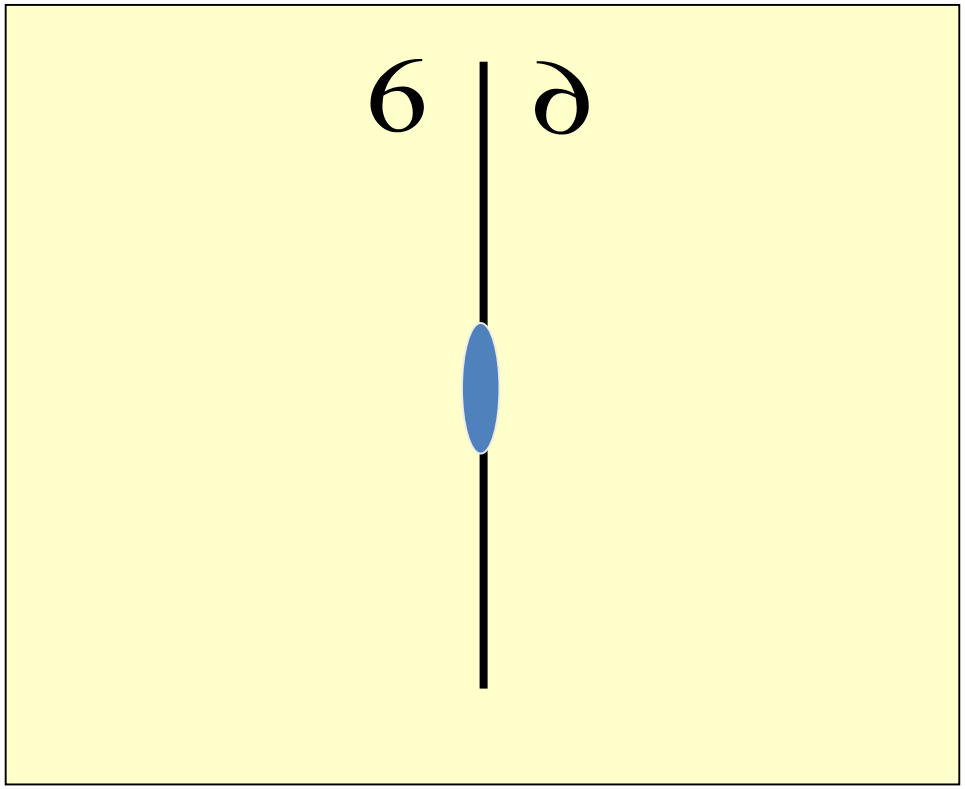


# 2-D Symmetry

Try combining a 2-fold rotation axis with a mirror

Step 1: reflect

(could do either step first)



# 2-D Symmetry

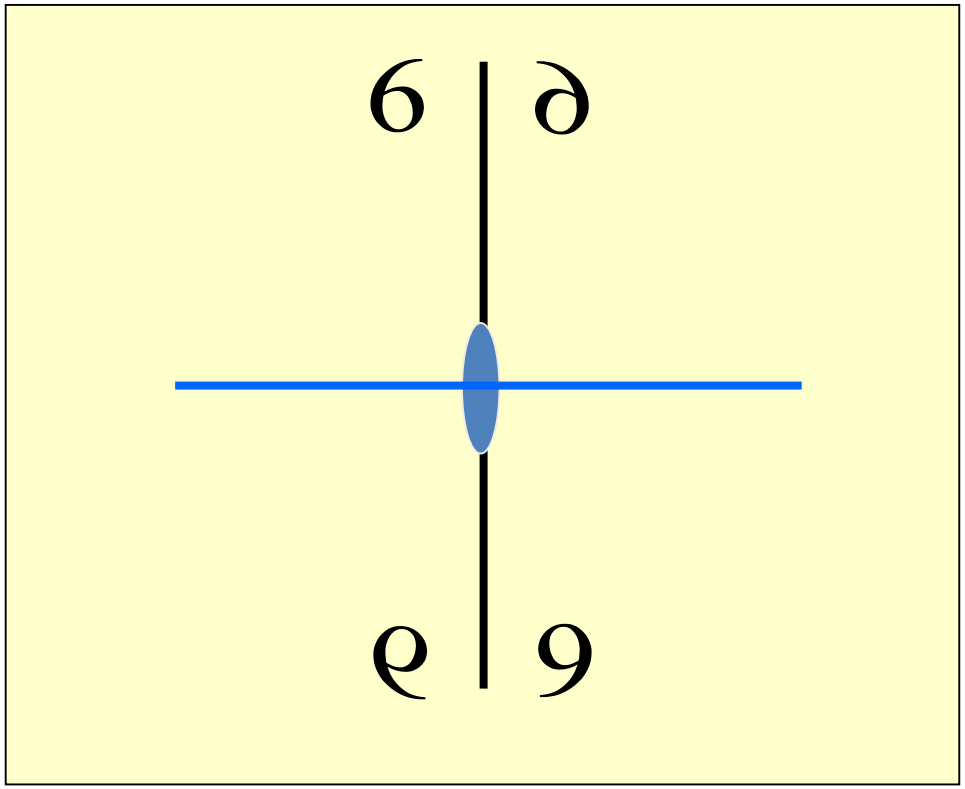
Try combining a 2-fold rotation axis with a mirror

Step 1: reflect

Step 2: rotate (everything)

Is that all??

No! A second mirror is required



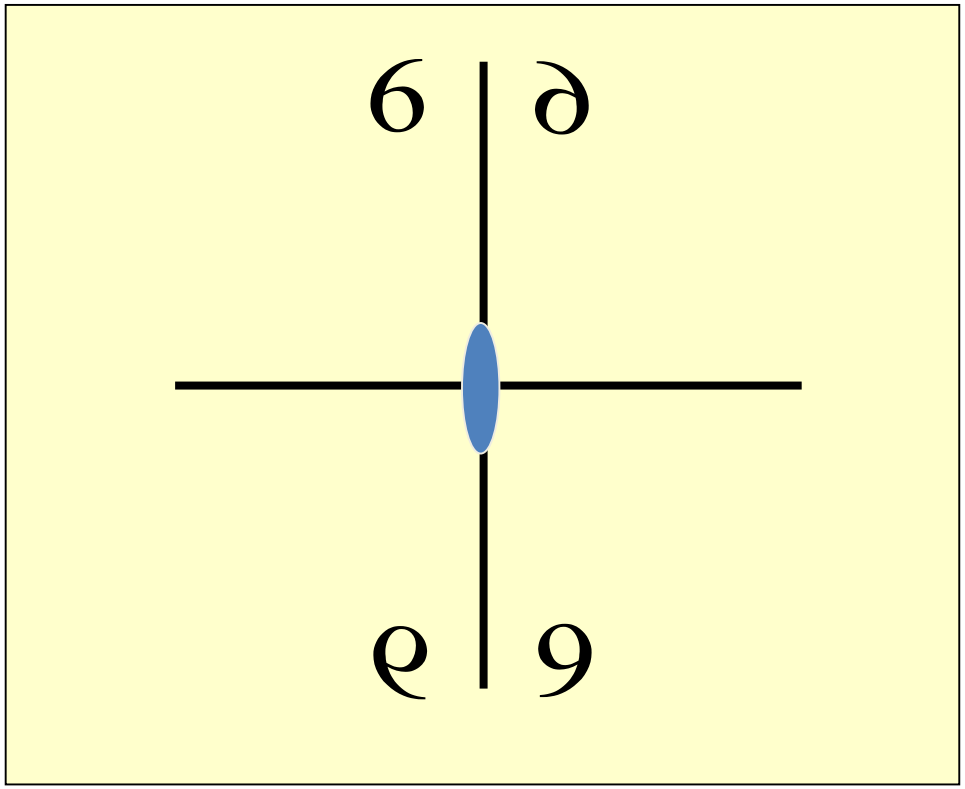
# 2-D Symmetry

Try combining a 2-fold rotation axis with a mirror

The result is Point Group 2mm

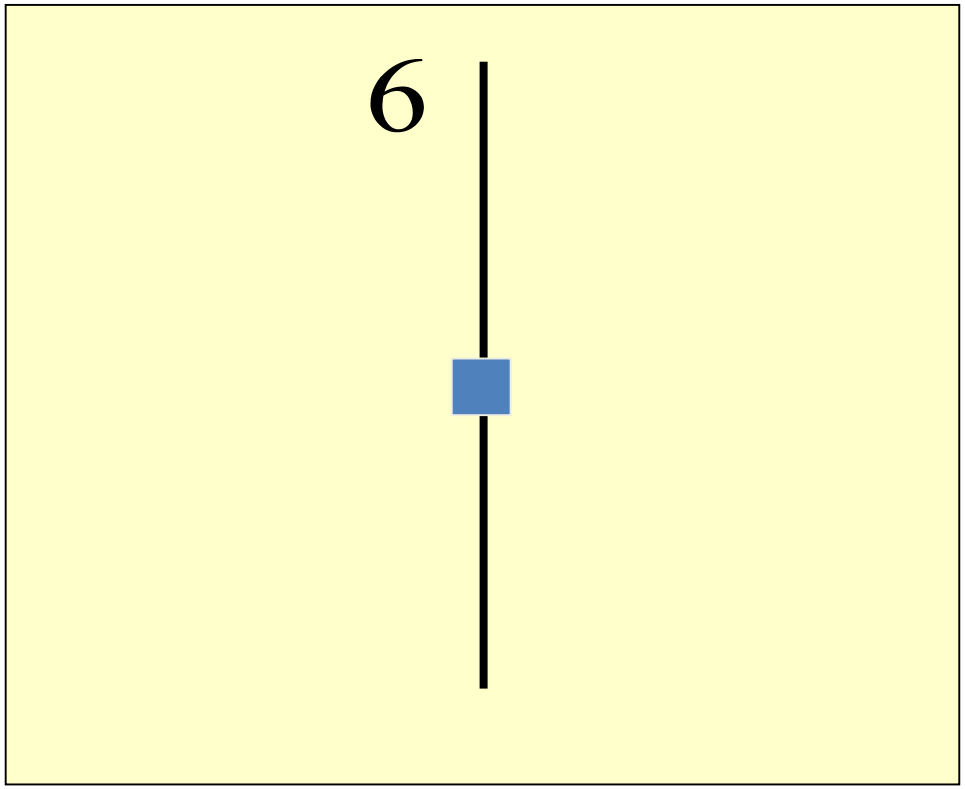
"2mm" indicates 2 mirrors

The mirrors are different
(not equivalent by symmetry)



# 2-D Symmetry

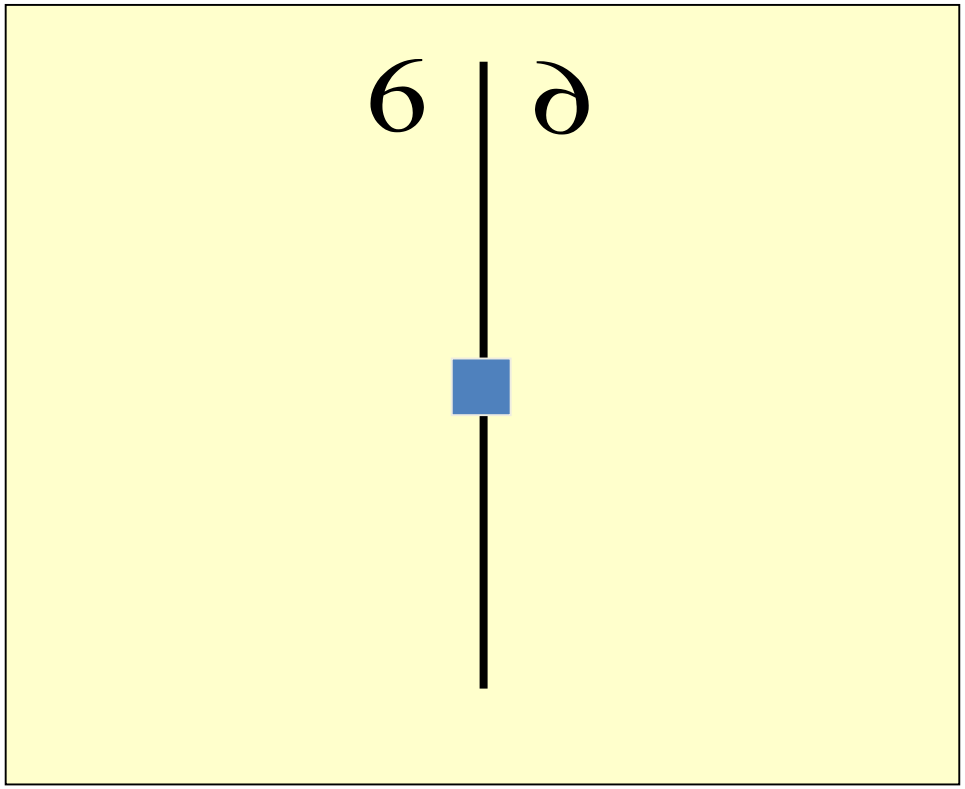
Now try combining a 4-fold rotation axis with a mirror



# 2-D Symmetry

Now try combining a 4-fold rotation axis with a mirror

Step 1: reflect

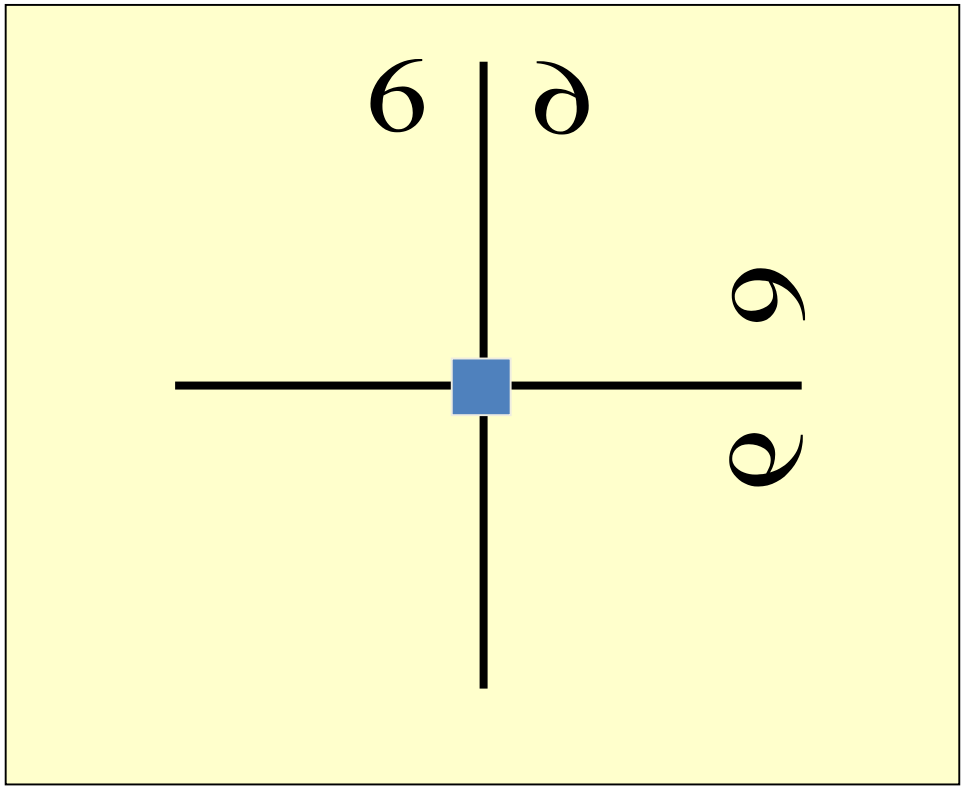


# 2-D Symmetry

Now try combining a 4-fold rotation axis with a mirror

Step 1: reflect

Step 2: rotate 1

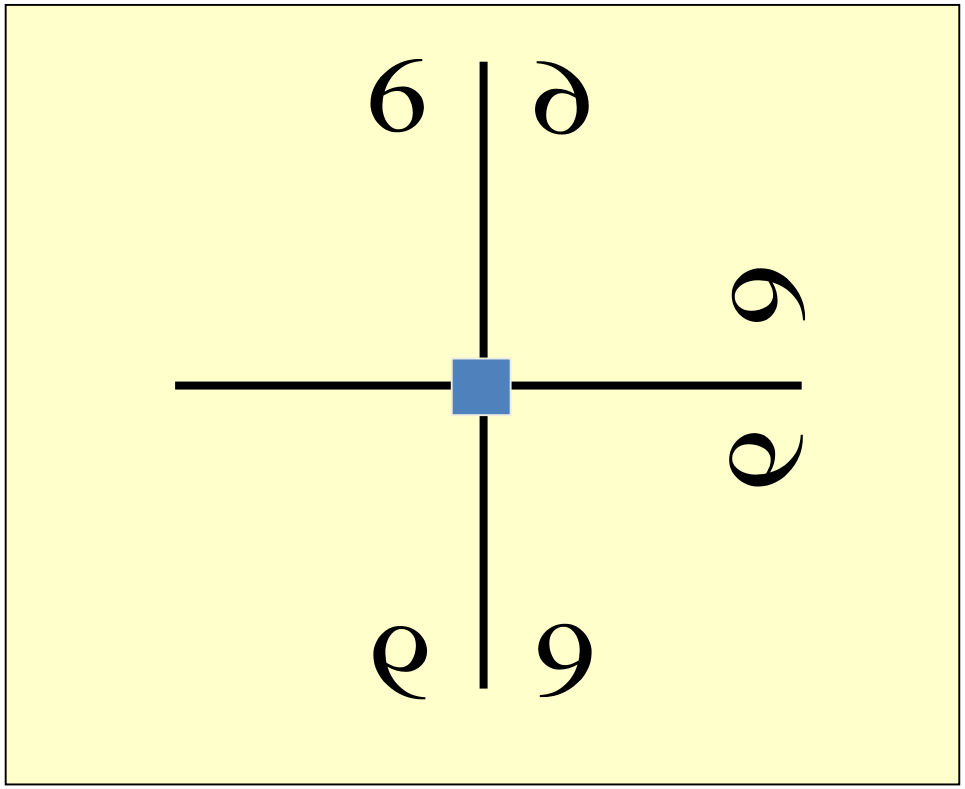


# 2-D Symmetry

Now try combining a 4-fold rotation axis with a mirror

Step 1: reflect

Step 2: rotate 2

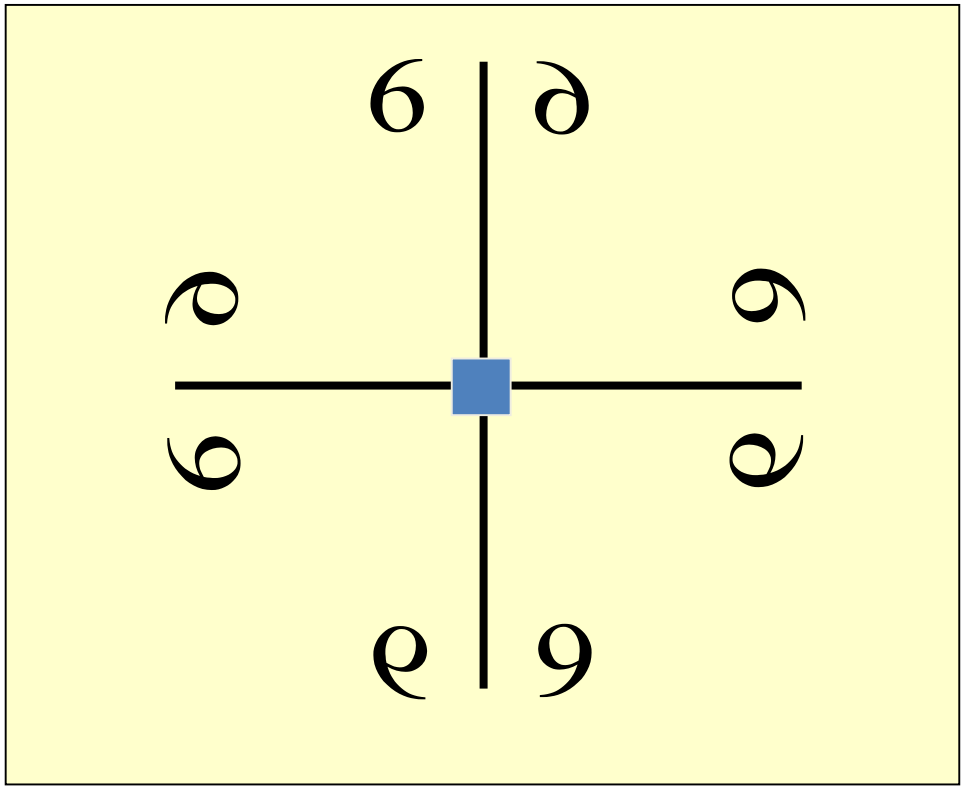


# 2-D Symmetry

Now try combining a 4-fold rotation axis with a mirror

Step 1: reflect

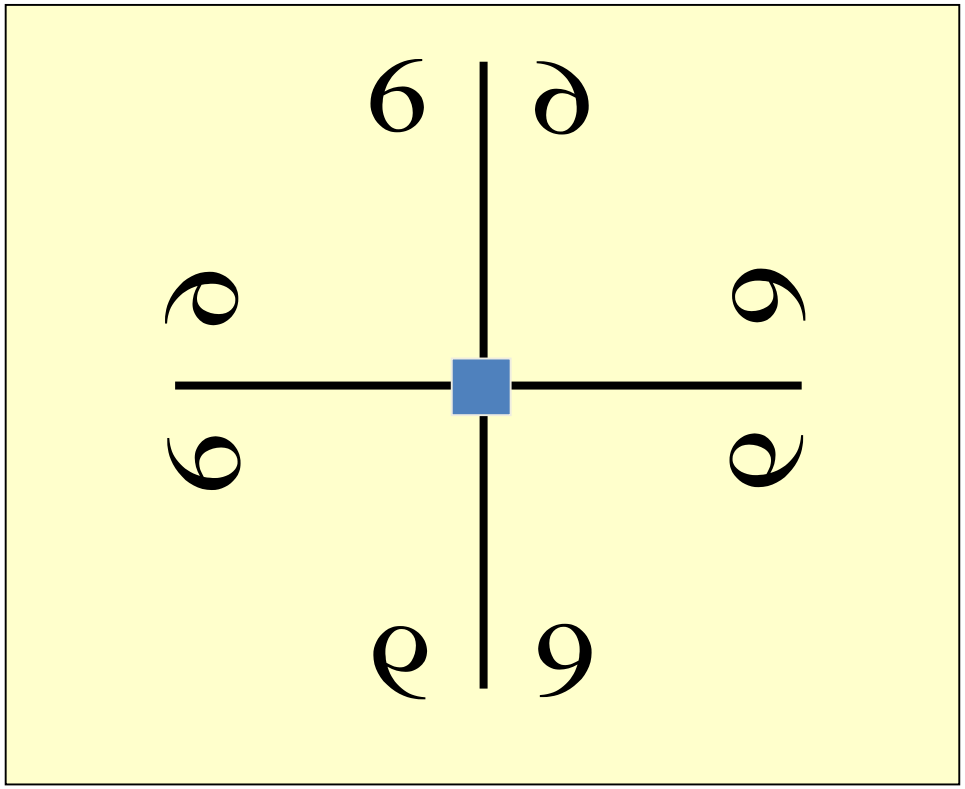
Step 2: rotate 3



# 2-D Symmetry

Now try combining a 4-fold rotation axis with a mirror

Any other elements?

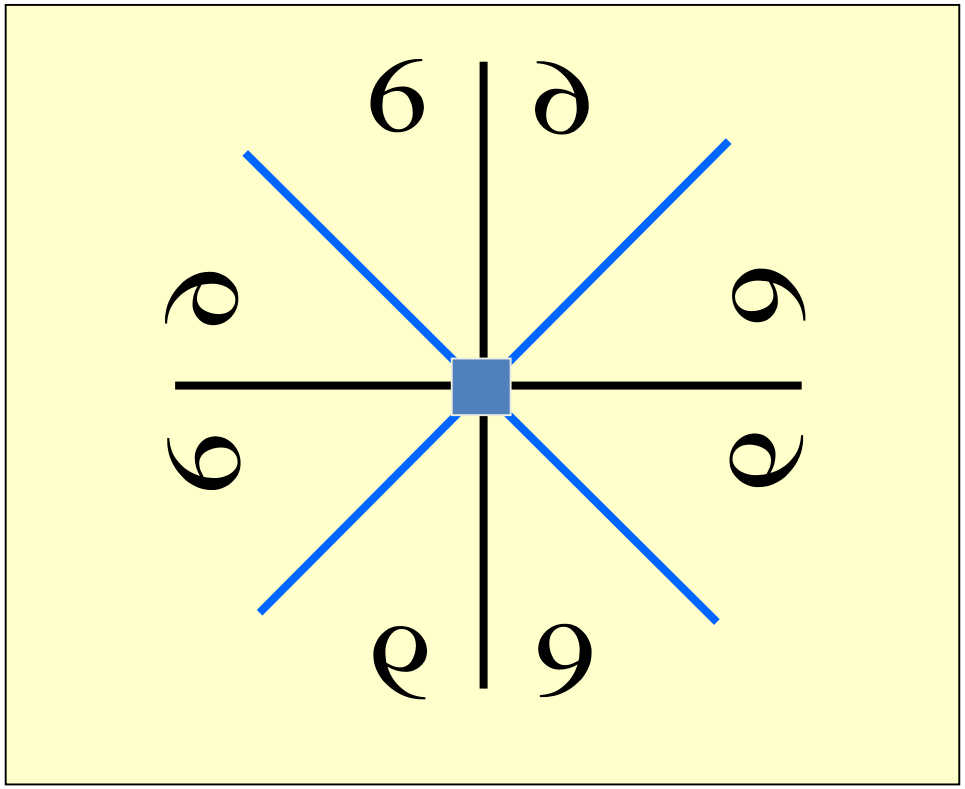


# 2-D Symmetry

Now try combining a 4-fold rotation axis with a mirror

Any other elements?

Yes, two more mirrors



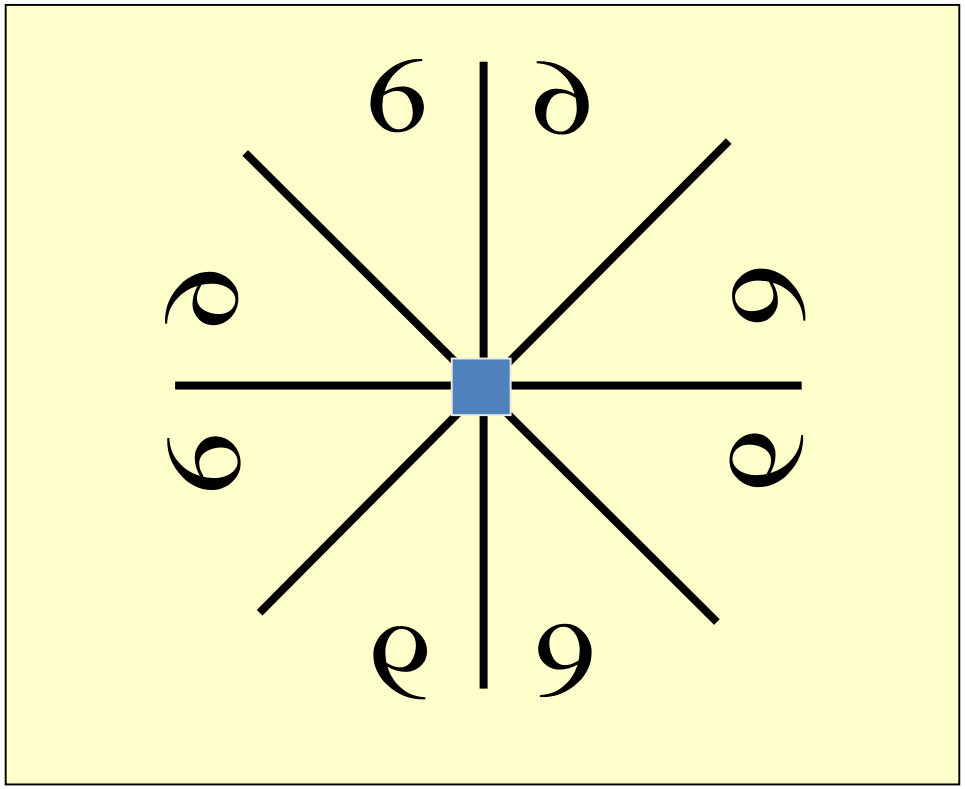
# 2-D Symmetry

Now try combining a 4-fold rotation axis with a mirror

Any other elements?

Yes, two more mirrors

Point group name??



# 2-D Symmetry

Now try combining a 4-fold rotation axis with a mirror

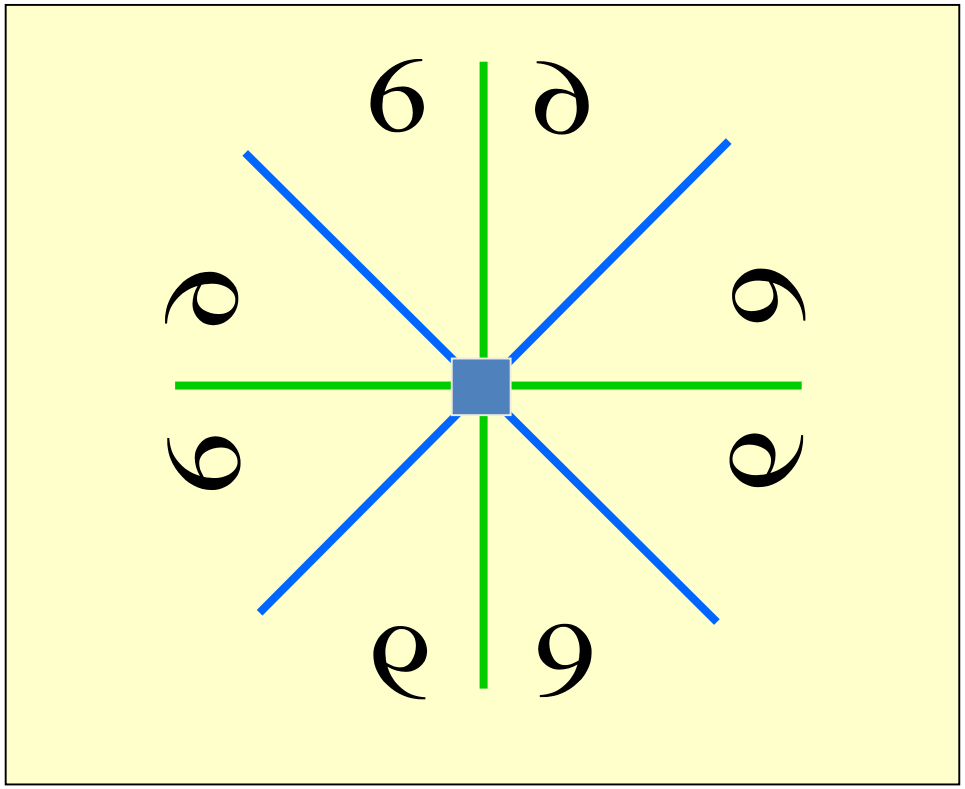
Any other elements?

Yes, two more mirrors

Point group name??

4mm

Why not 4mmmm?



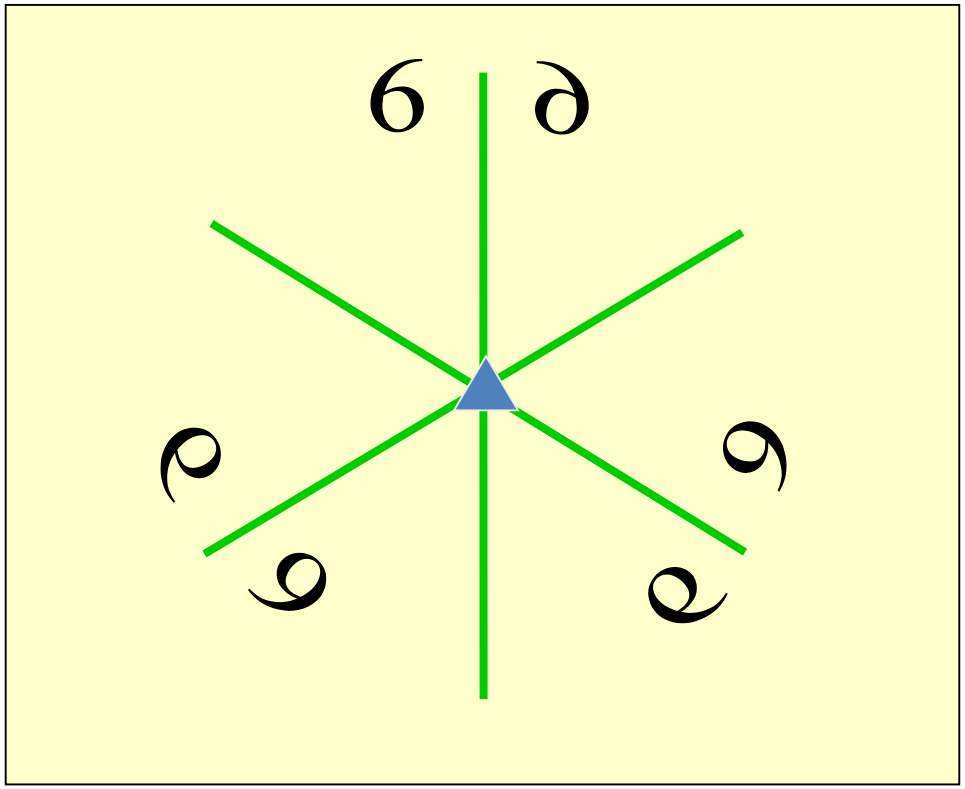
# 2-D Symmetry

3-fold rotation axis with a mirror

Guess the point group.

Point group 3m

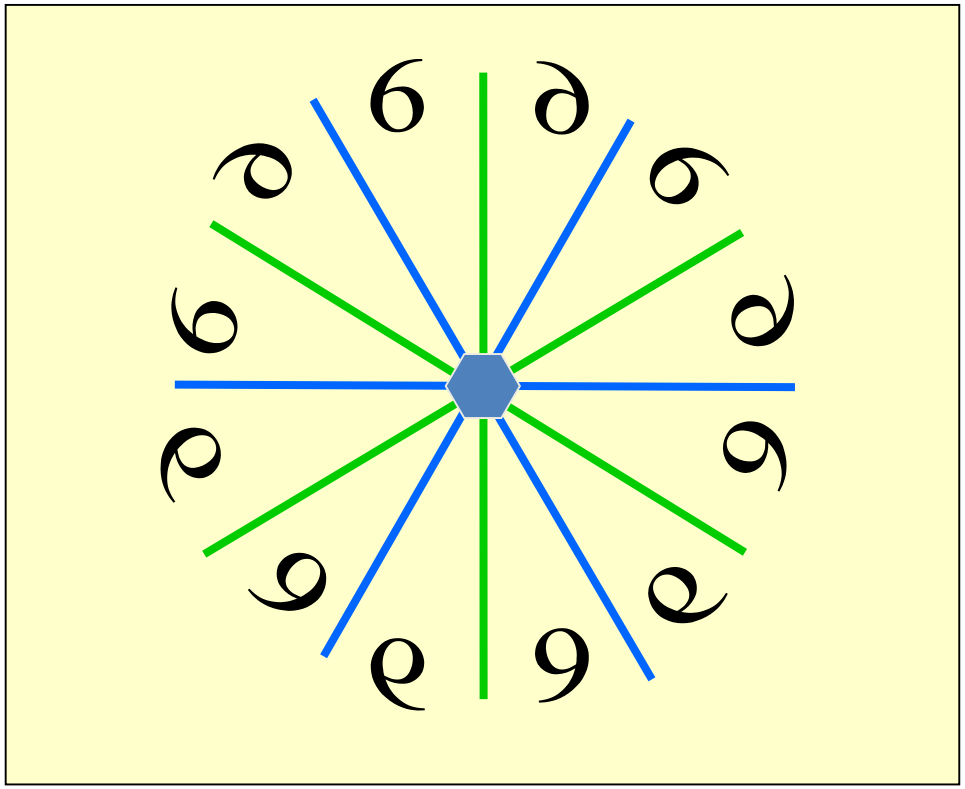
Why not 3mmm?



# 2-D Symmetry

6-fold rotation axis with a mirror creates ???

point group 6mm



# 2-D Symmetry

The original 6 elements plus the 4 combinations creates 10 possible 2-D Point Groups:

1 2 3 4 6 m 2mm 3m 4mm 6mm

*Any* repeating 2-D pattern of objects surrounding a point must conform to one of these groups

For example, surfaces or interfaces sometimes have a different symmetry than the bulk.

# 2D vs 3D

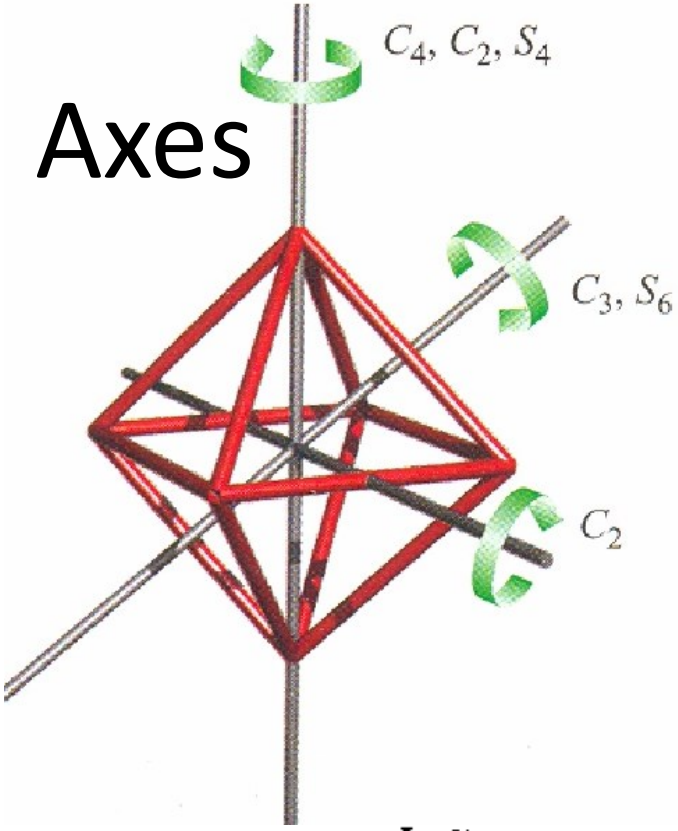
- In 2D, have 6 unique symmetry operations:

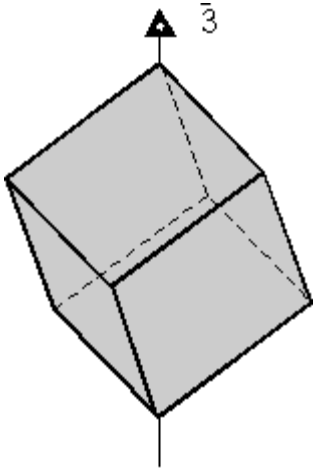
  1 2 3 4 6 m (inversion not unique)

- 2D has 10 point groups: 1 2 3 4 6 m 2mm 3m 4mm 6mm
- In 3D, inversion is unique
- Now 7 symmetry operations (sort of)
- **Every crystal** described by a combination of point-symmetry elements

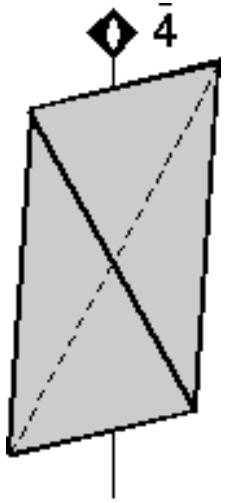
# Rotation-inversion Axes

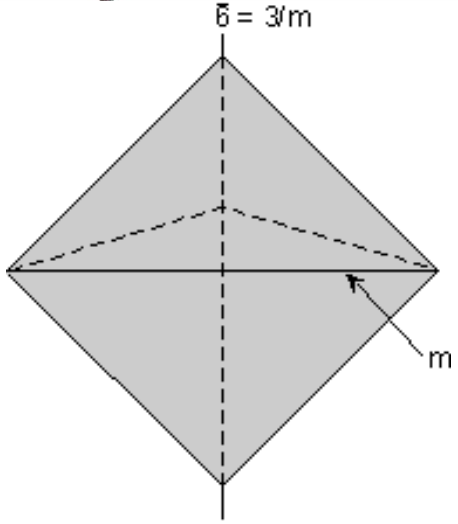
- Combination of simultaneous rotation and inversion results in new symmetry element
- International symbol representation: $\bar{2}, \bar{3}, \bar{4}, \bar{6}$




This operation involves a rotation by (360/3) ° followed by an inversion through the center of the object.



# 3-D Symmetry

We now have 10 unique 3-D symmetry operations:

1 2 3 4 6 *i* m $\bar{3}$ $\bar{4}$ $\bar{6}$

Combinations of these elements are also possible

A complete analysis of symmetry about a point in space requires that we try all possible combinations of these symmetry elements

# Symmetry operations

## Various Terminology

| | |
|---|---|
| $E$ | The identity transformation |
| $C_n$ | Rotation (clockwise) through an angle of $2\pi/n$ radians, where $n$ is an integer. |

| Element | Schoenflies | Hermann-Mauguin | Operation |
|---|---|---|---|
| Rotation axis | $C_n$ | n | n-fold rotation (360º/n) |
| Identity | E | 1 | nothing |
| Plane of symmetry | $\sigma$ | m | Reflection |
| Center of symmetry | $i$ | -1 | Inversion |
| Improper rotation axis | $S_n$ | - | n-fold rotation + reflection |
| Rotary Inversion axis | - | -n or “n bar” | n-fold rotation + inversion |

http://www.staff.ncl.ac.uk/j.p.goss/symmetry/CharacterTables.html

# 3-D Symmetry

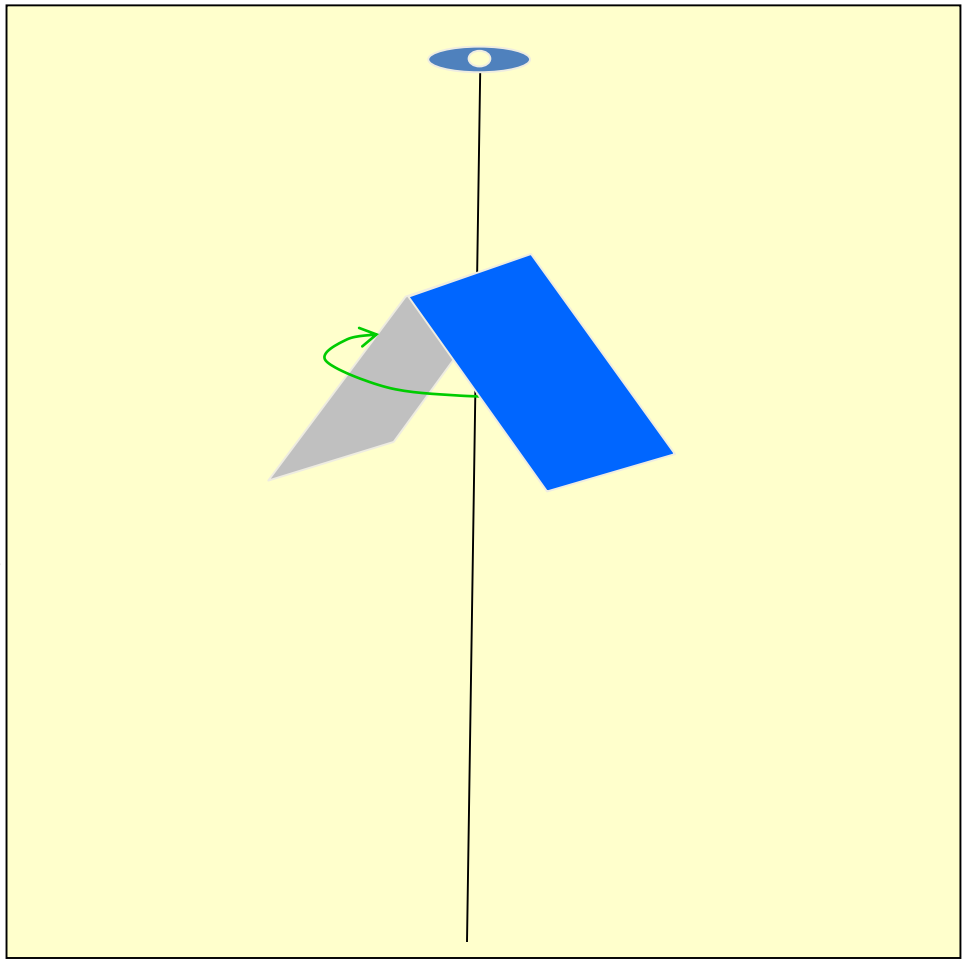
New Symmetry Elements

4. Rotoinversion

b. 2-fold rotoinversion ( $\bar{2}$ )

Step 1: rotate 360/2

Note: this is a temporary step, the intermediate motif element does not exist in the final pattern



# 3-D Symmetry

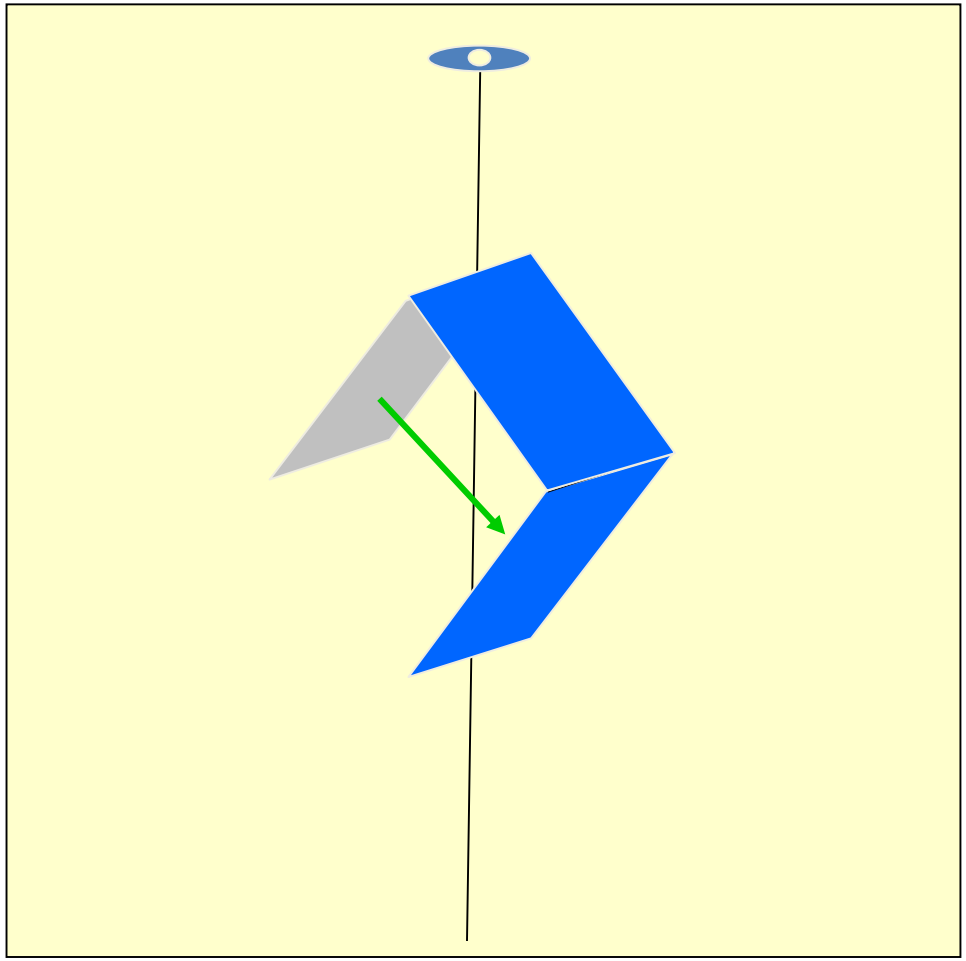
New Symmetry Elements

4. Rotoinversion

b. 2-fold rotoinversion ( $\bar{2}$ )

Step 1: rotate 360/2

Step 2: invert



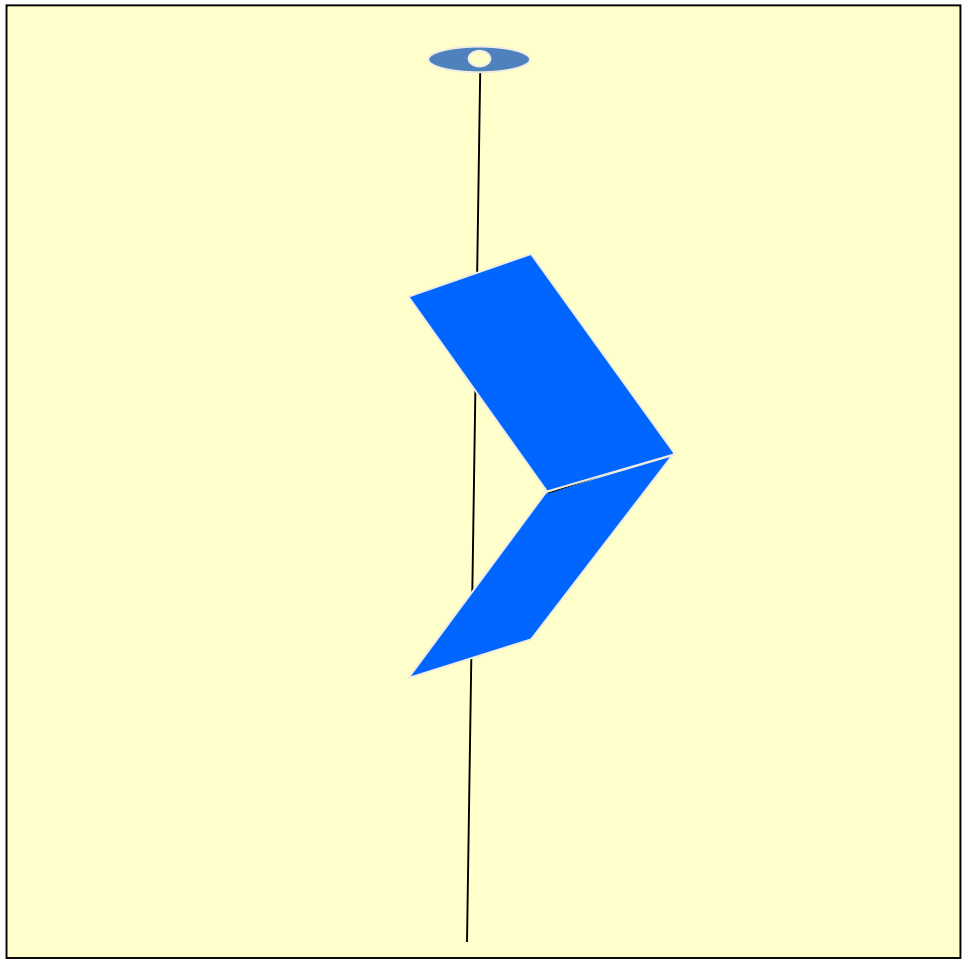
# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

b. 2-fold rotoinversion ( $\bar{2}$ )

The result:



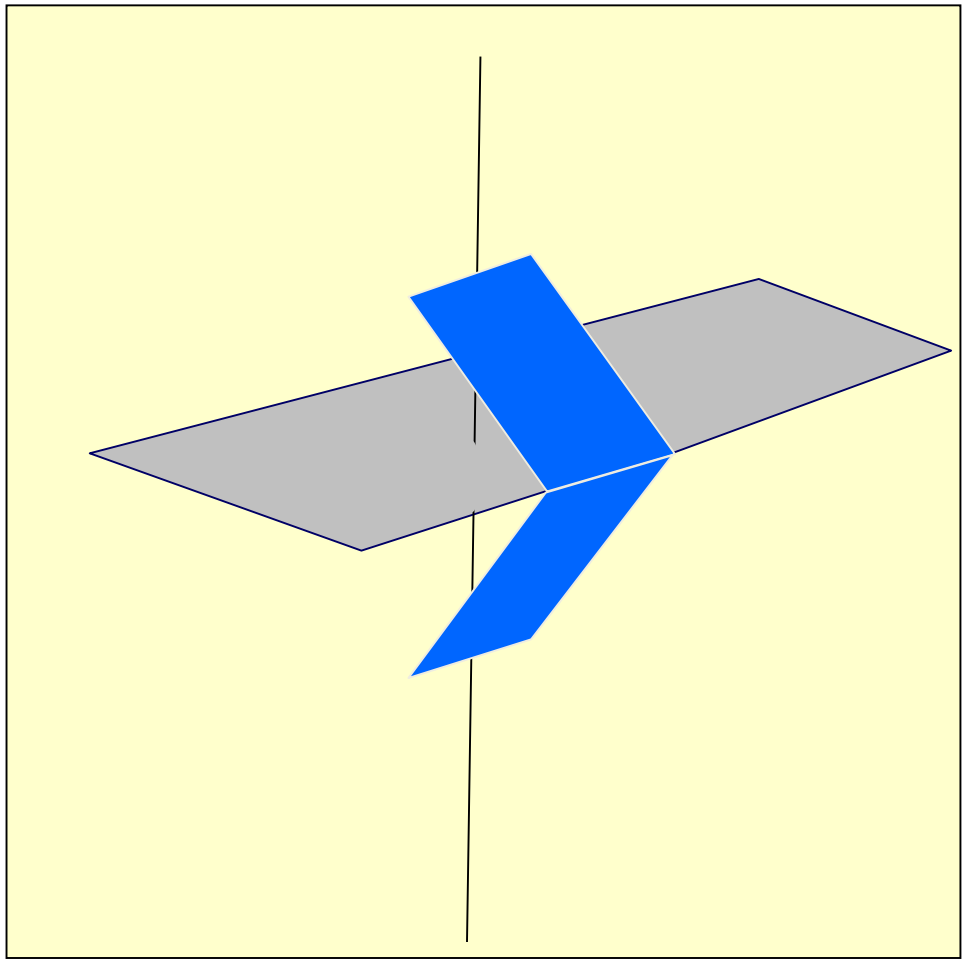
# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

b. 2-fold rotoinversion ( $\bar{2}$ )

This is the same as ***m***, so not a new operation

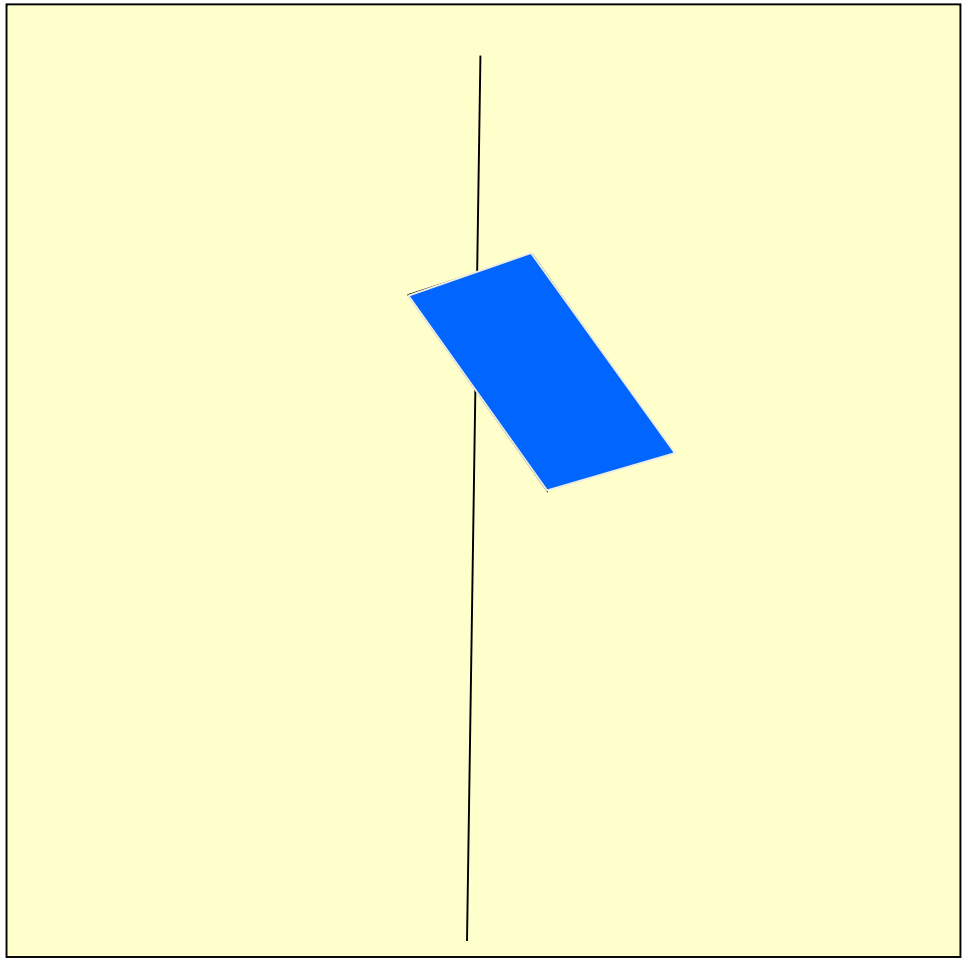


# 3-D Symmetry

New 3-D Symmetry Elements

4. Rotoinversion

   a. 1-fold rotoinversion ( $\bar{1}$ )



# 3-D Symmetry

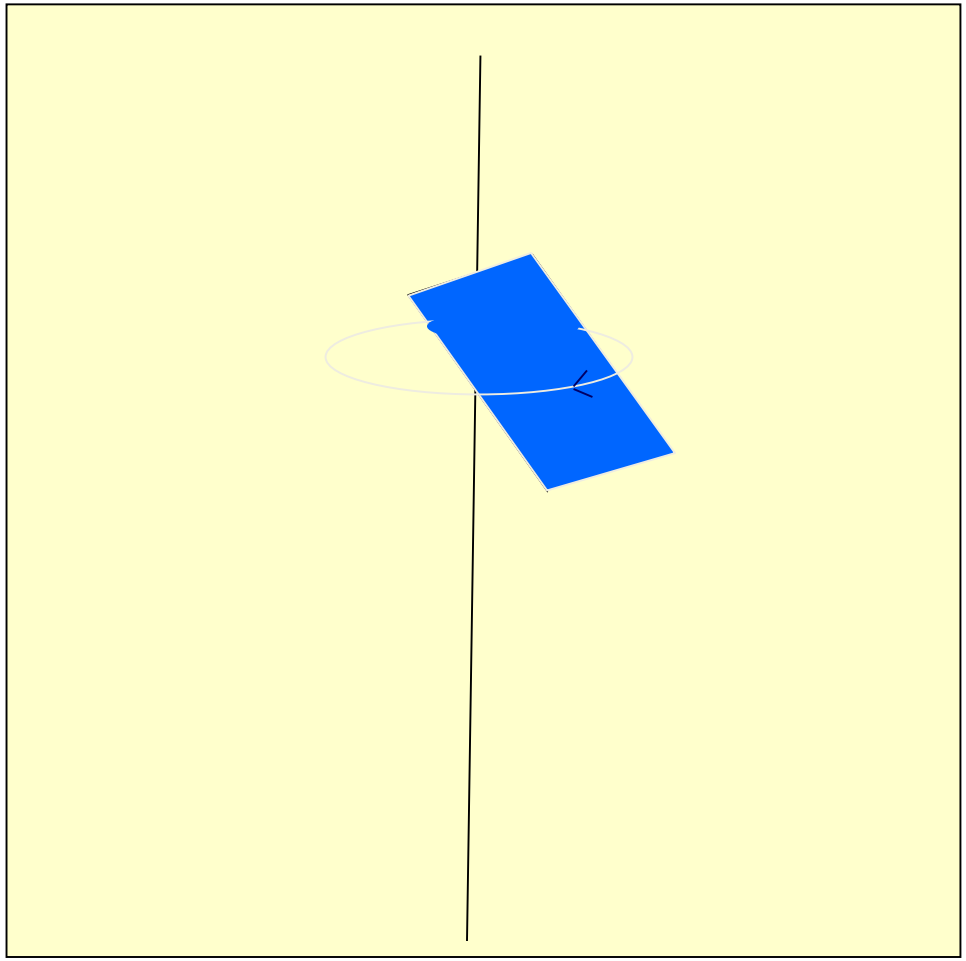
New 3-D Symmetry Elements

4. Rotoinversion

a. 1-fold rotoinversion ( $\bar{1}$ )

Step 1: rotate 360/1

(identity)



# 3-D Symmetry

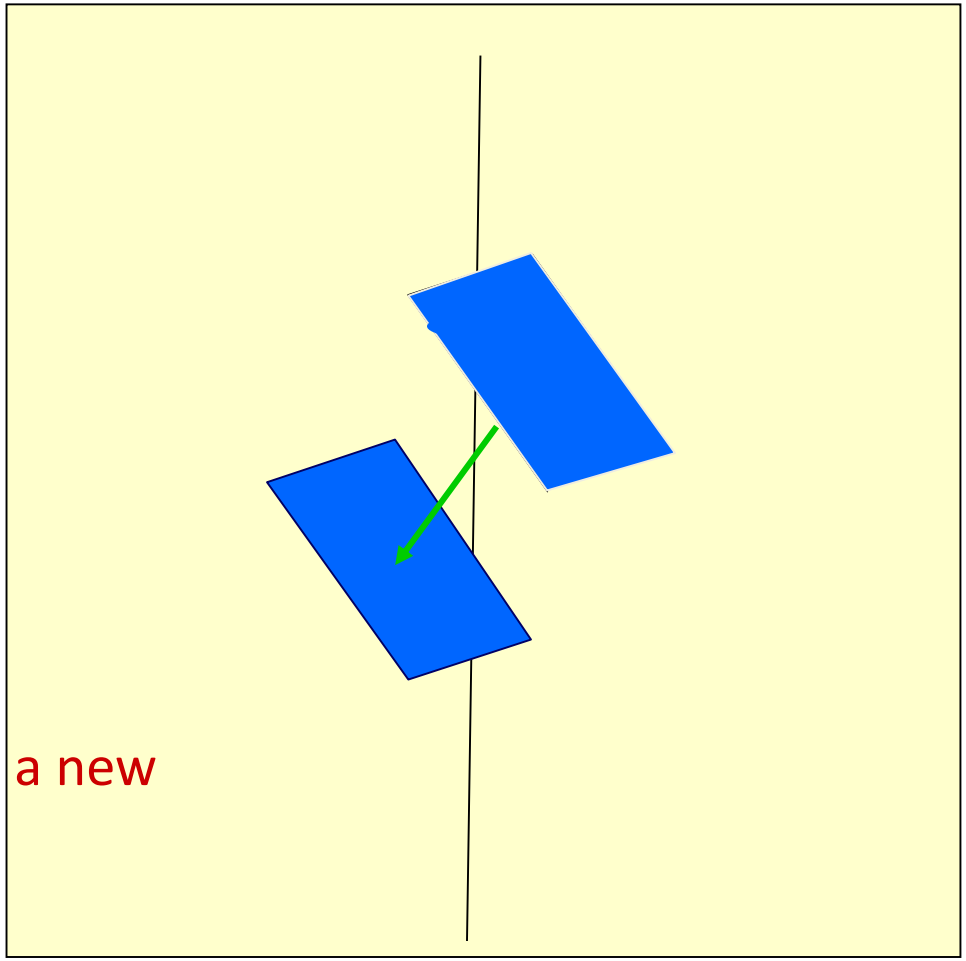
New 3-D Symmetry Elements

4. Rotoinversion

a. 1-fold rotoinversion ( 1 )

Step 1: rotate 360/1 (identity)

Step 2: invert (x,y,z) -> (-x,-y,-z)



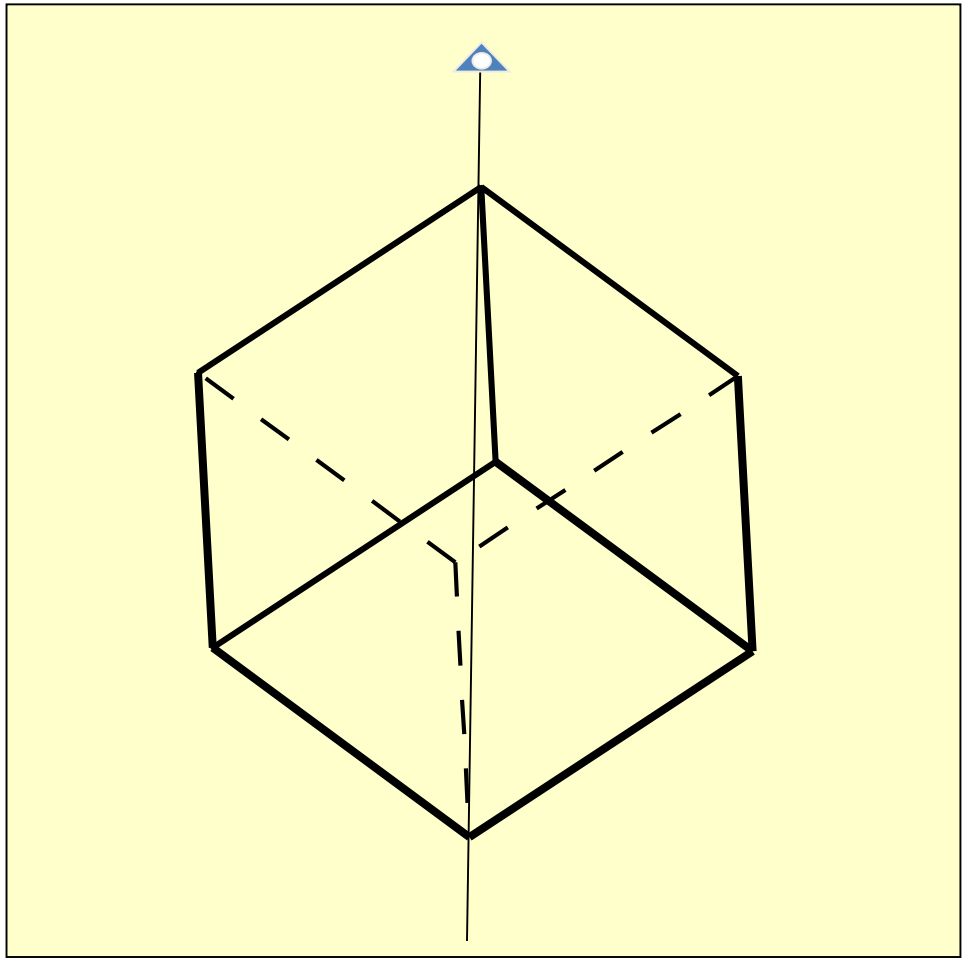
This is the same as ***i***, so also not a new operation

# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

c. 3-fold rotoinversion ( $\bar{3}$ )



# 3-D Symmetry

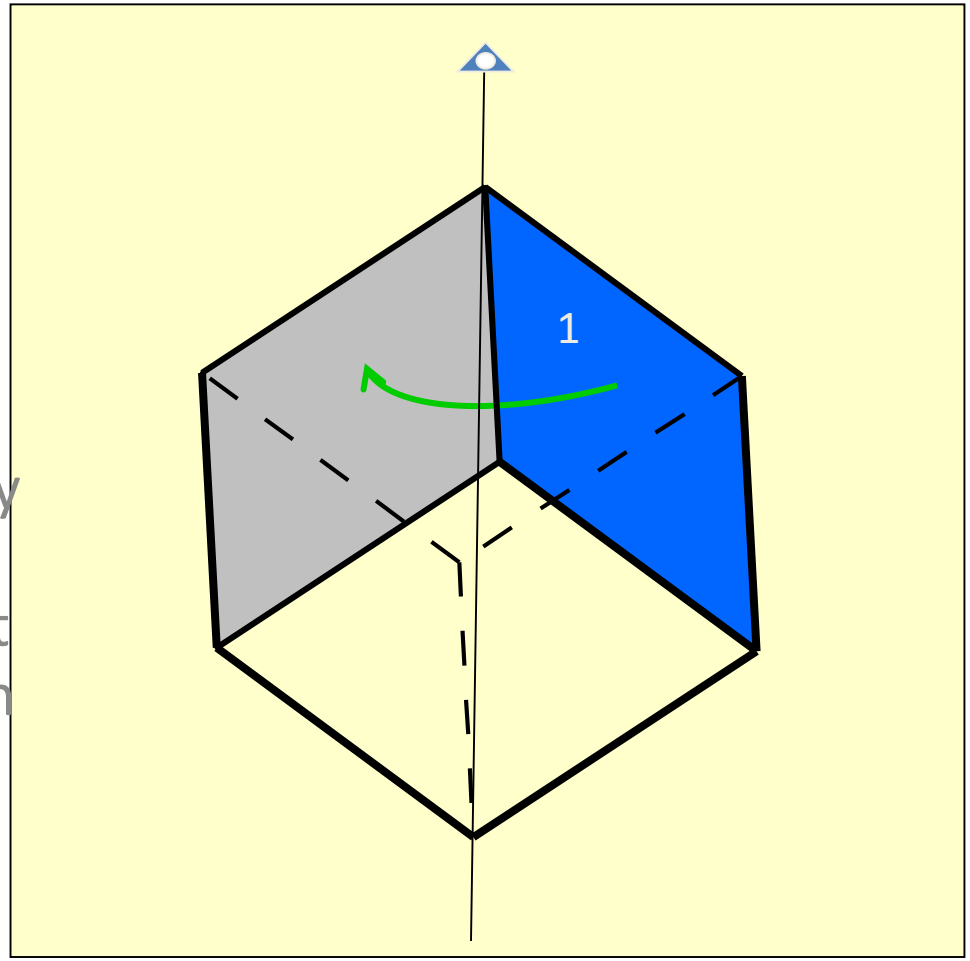
New Symmetry Elements

4. Rotoinversion

c. 3-fold rotoinversion ( $\bar{3}$ )

Step 1: rotate 360°/3

Again, this is a temporary step, the intermediate motif element does not exist in the final pattern



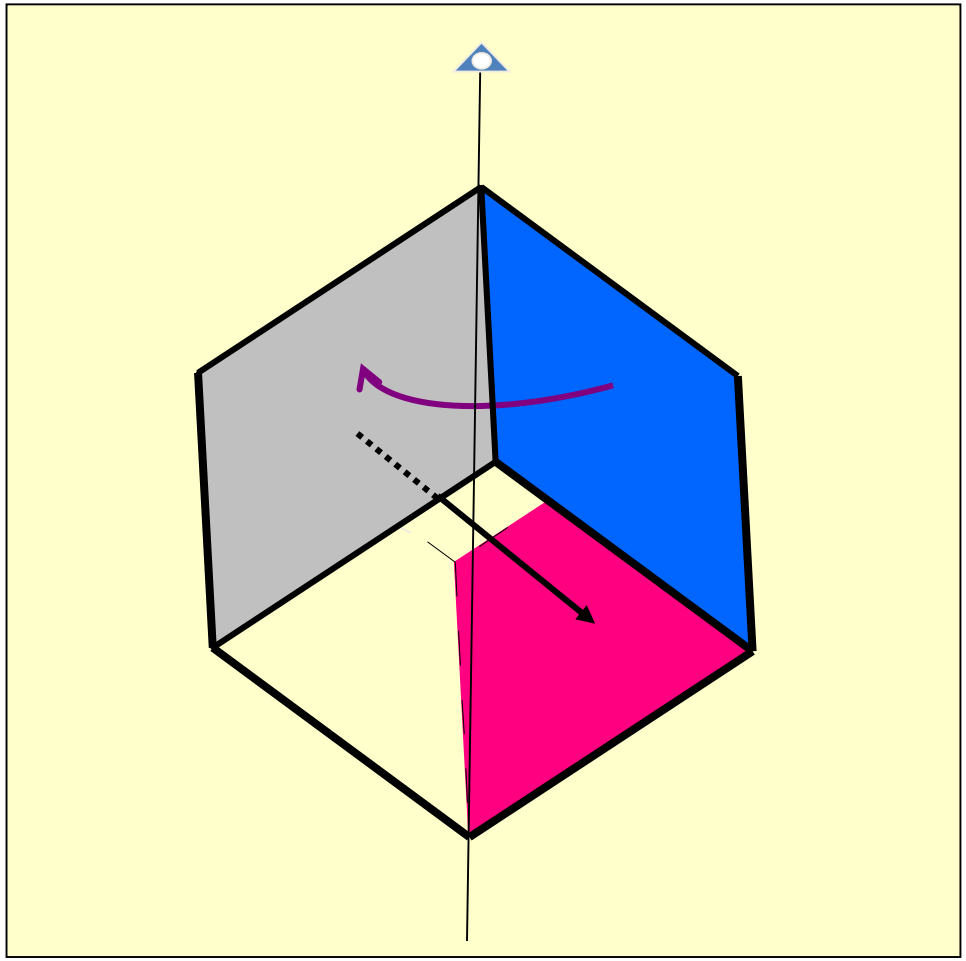
# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

c. 3-fold rotoinversion ( $\bar{3}$ )

Step 2: invert through center



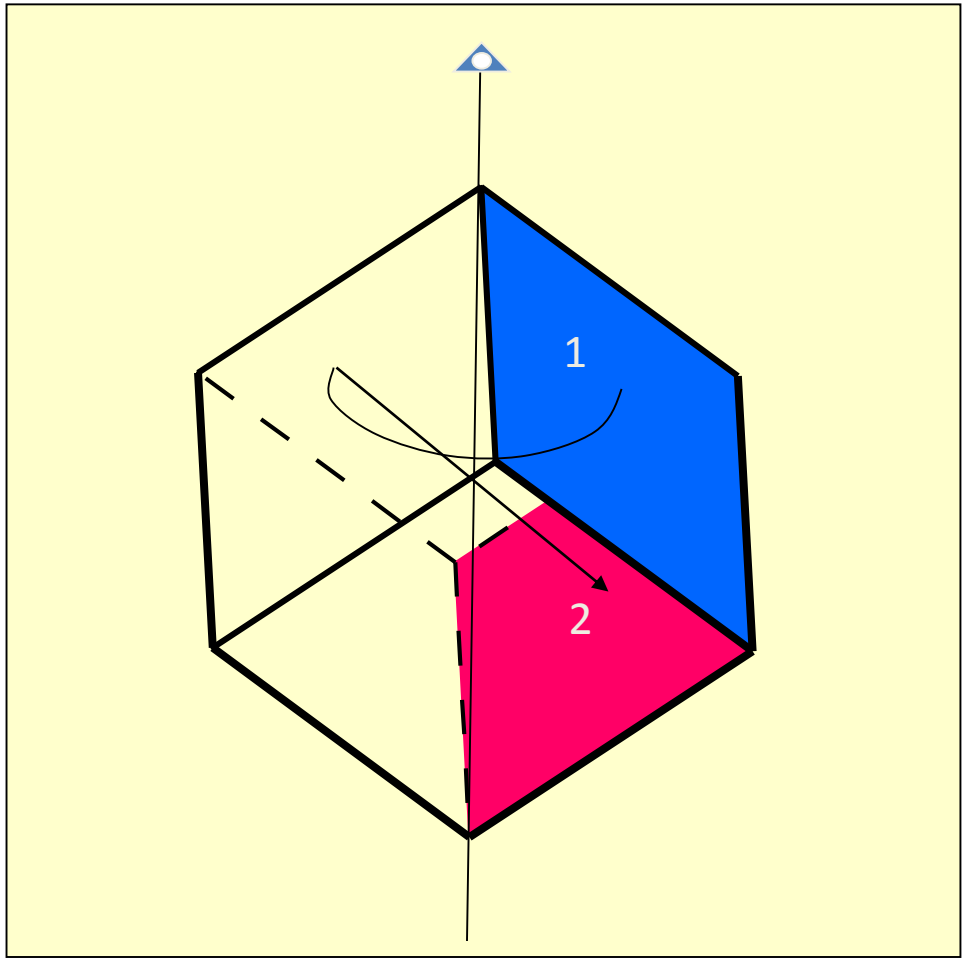
# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

c. 3-fold rotoinversion ( $\bar{3}$ )

Completion of the first sequence



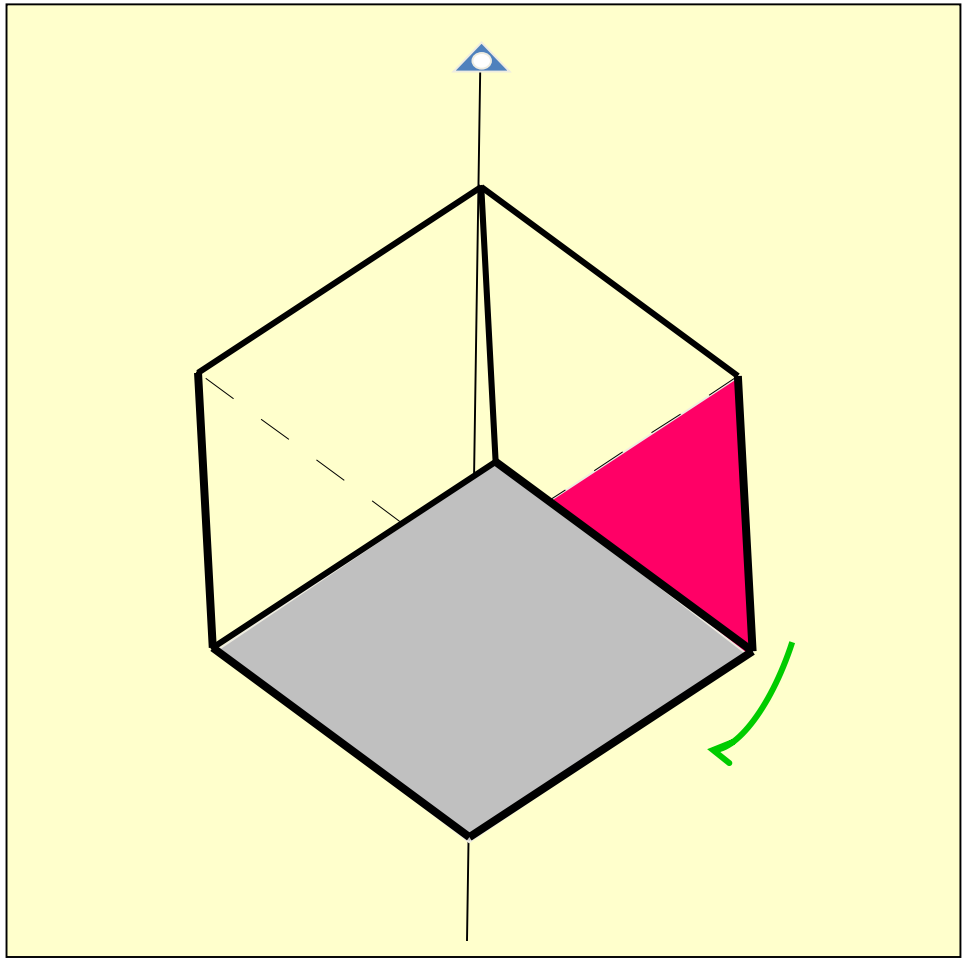
# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

c. 3-fold rotoinversion ( $\bar{3}$ )

Rotate another 360/3



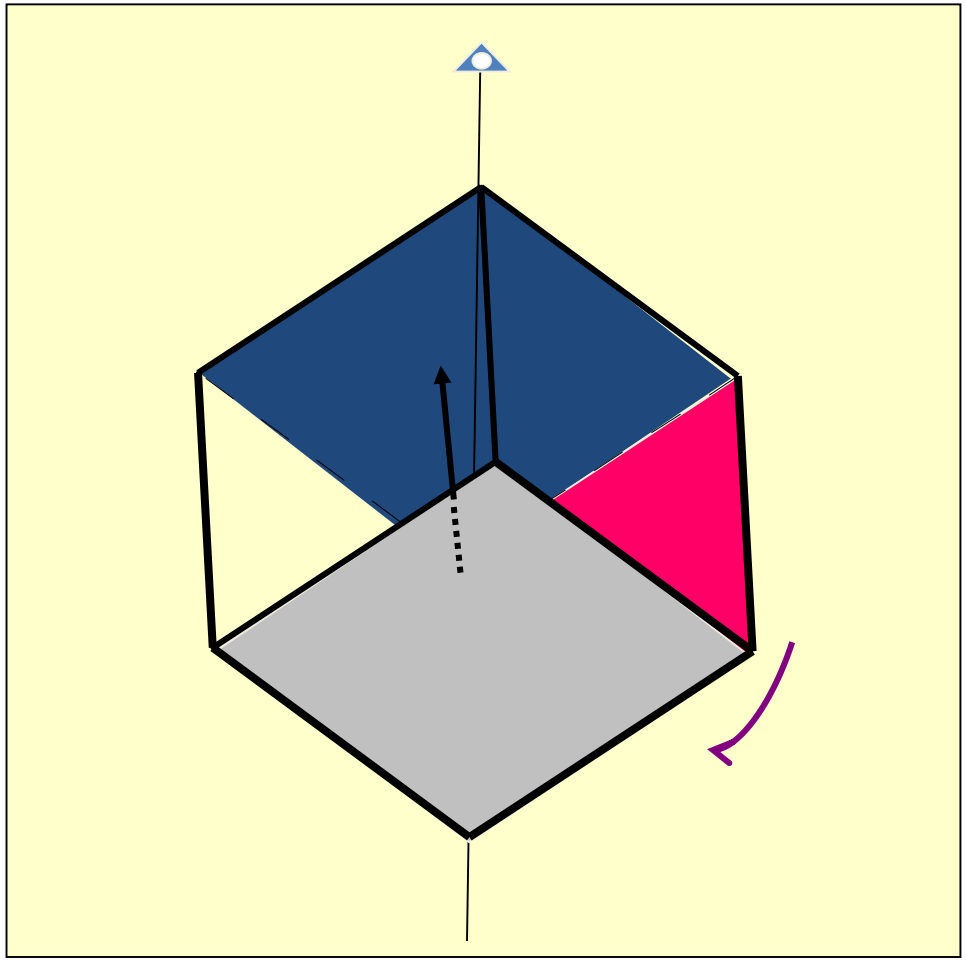
# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

c. 3-fold rotoinversion ( $\bar{3}$ )

Invert through center



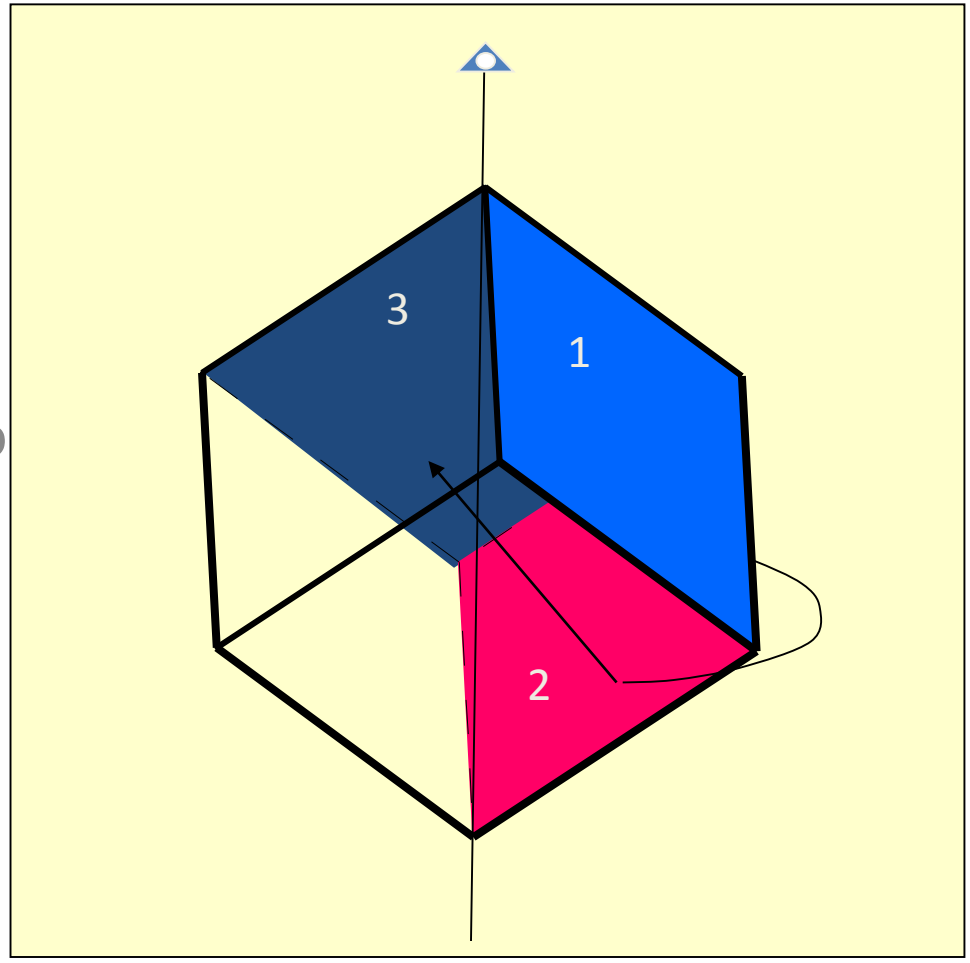
# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

c. 3-fold rotoinversion ( $\bar{3}$ )

Complete second step to create face 3



# 3-D Symmetry

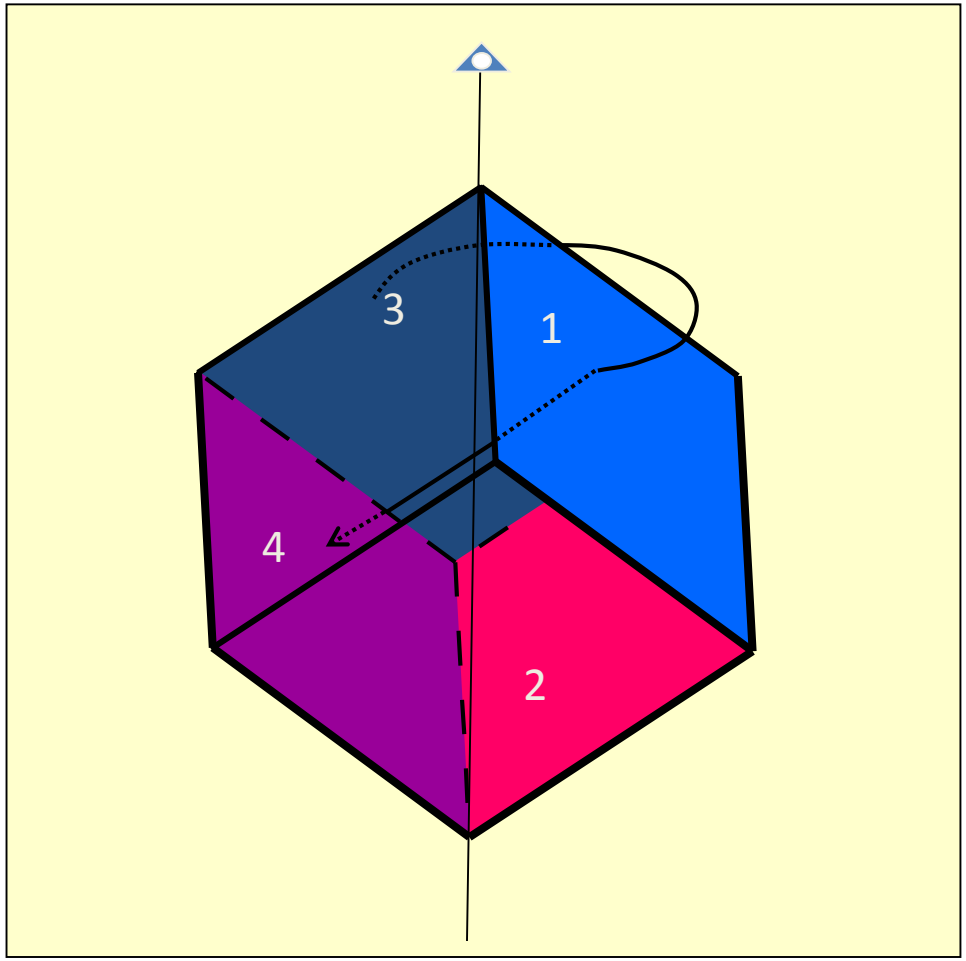
New Symmetry Elements

4. Rotoinversion

c. 3-fold rotoinversion ( $\bar{3}$ )

Third step creates face 4

$(3 \rightarrow (1) \rightarrow 4)$



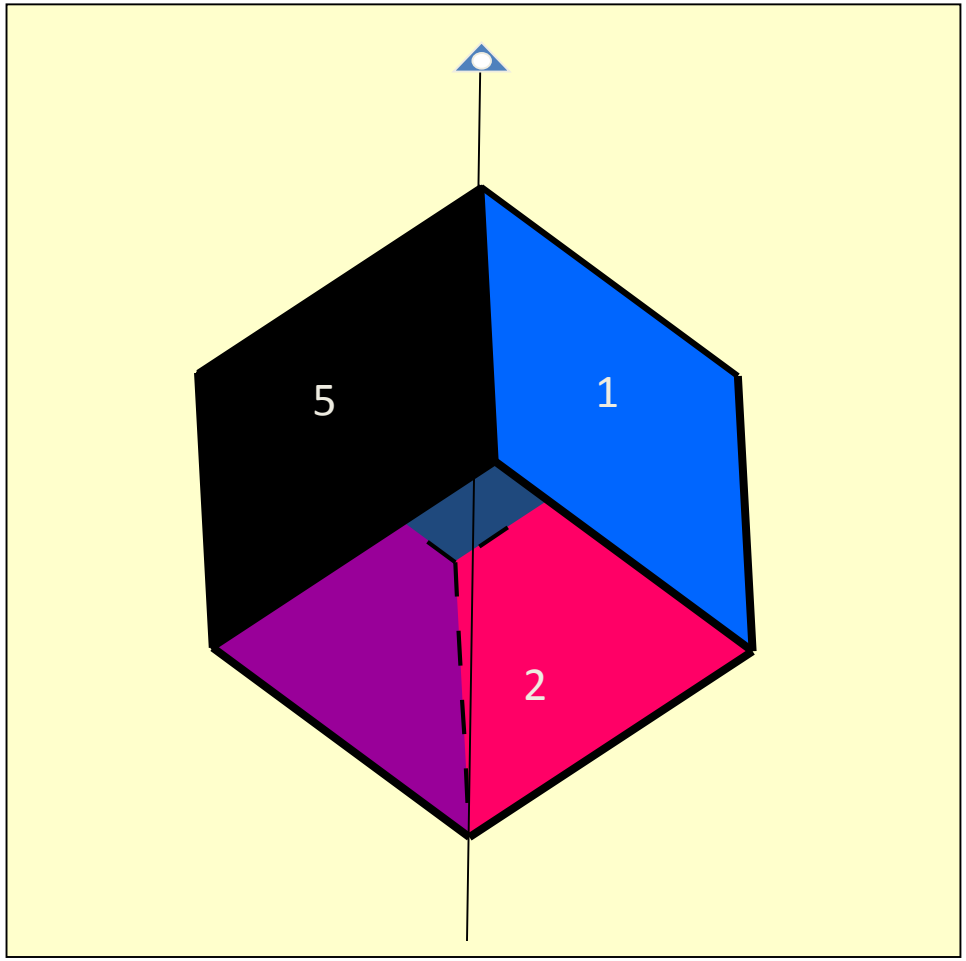
# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

c. 3-fold rotoinversion ( $\bar{3}$ )

Fourth step creates face 5 ($4 \rightarrow (2) \rightarrow 5$)



# 3-D Symmetry

New Symmetry Elements

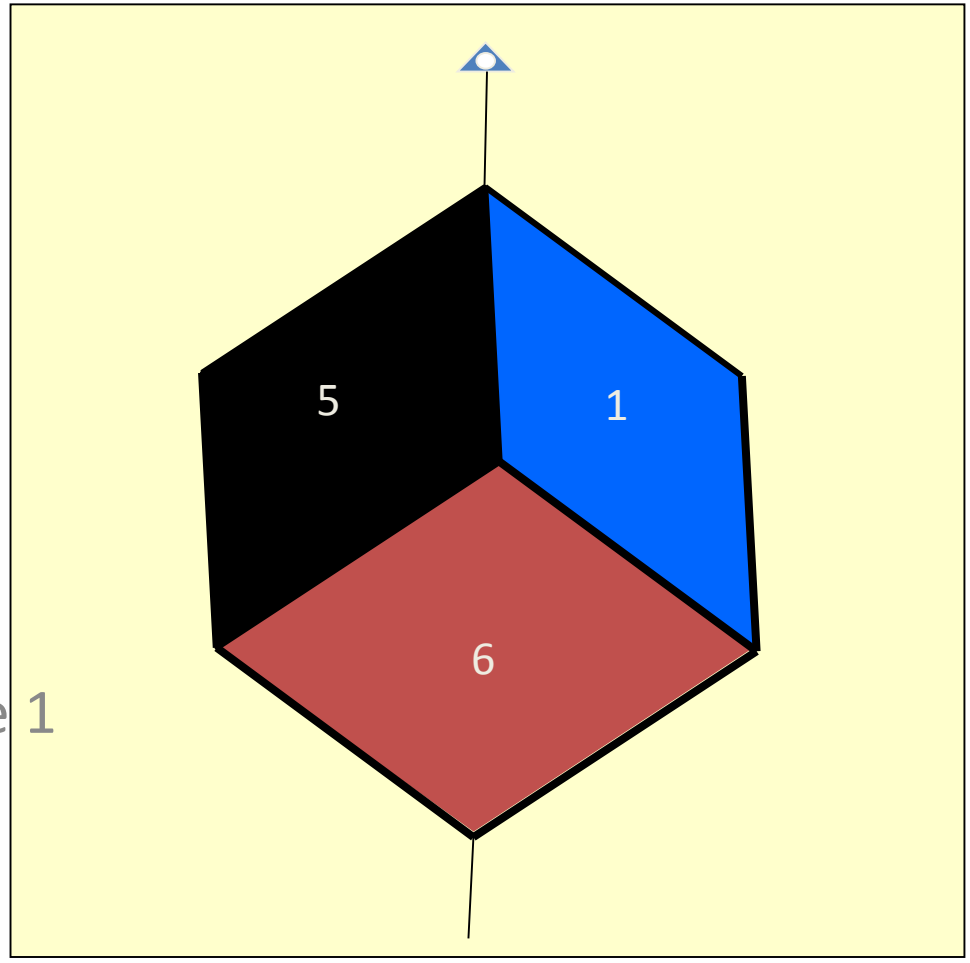
4. Rotoinversion

c. 3-fold rotoinversion ( $\bar{3}$ )

Fifth step creates face 6

$(5 \rightarrow (3) \rightarrow 6)$

Sixth step returns to face 1



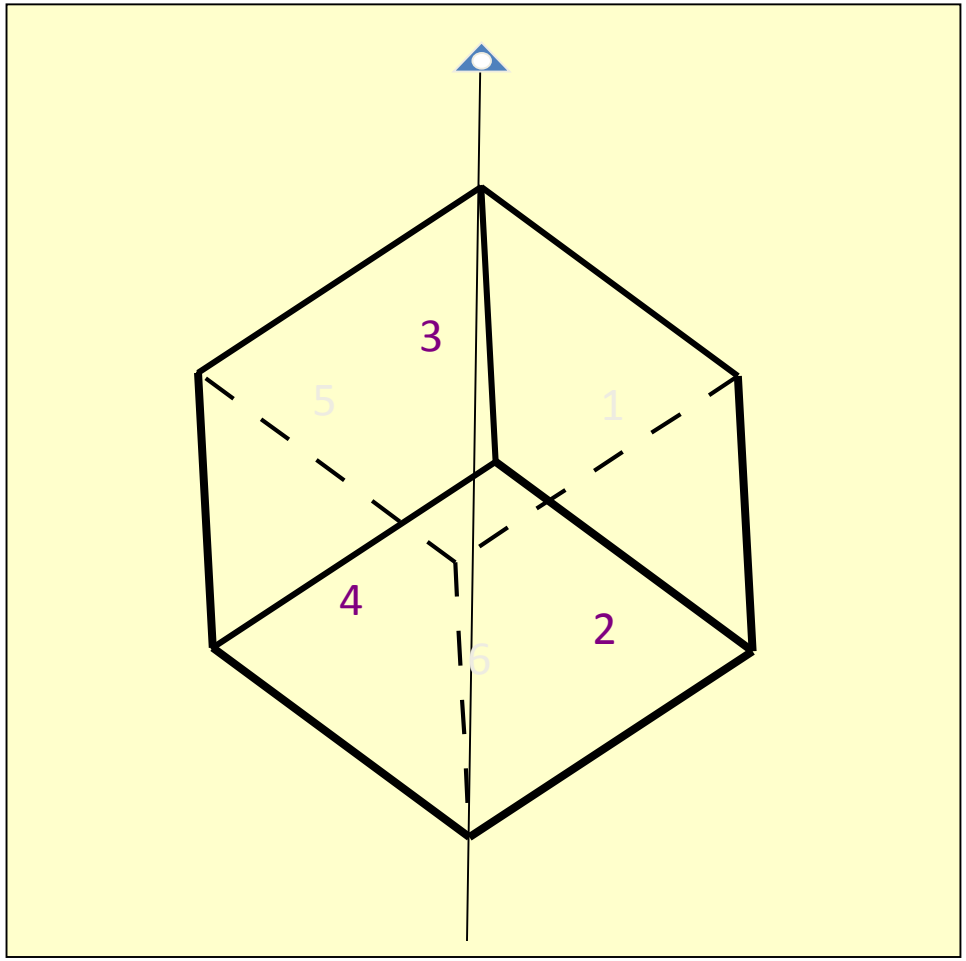
# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

c. 3-fold rotoinversion ( $\bar{3}$ )

This is unique

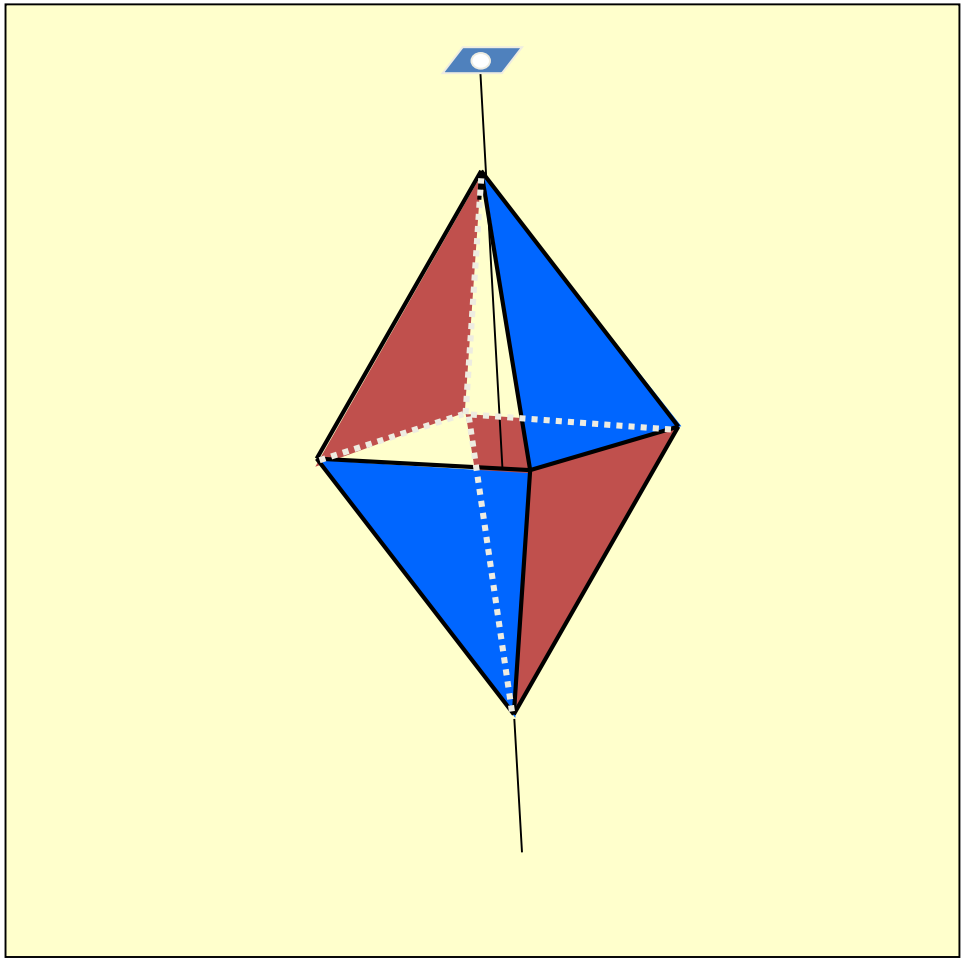


# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

d. 4-fold rotoinversion ( $\bar{4}$ )

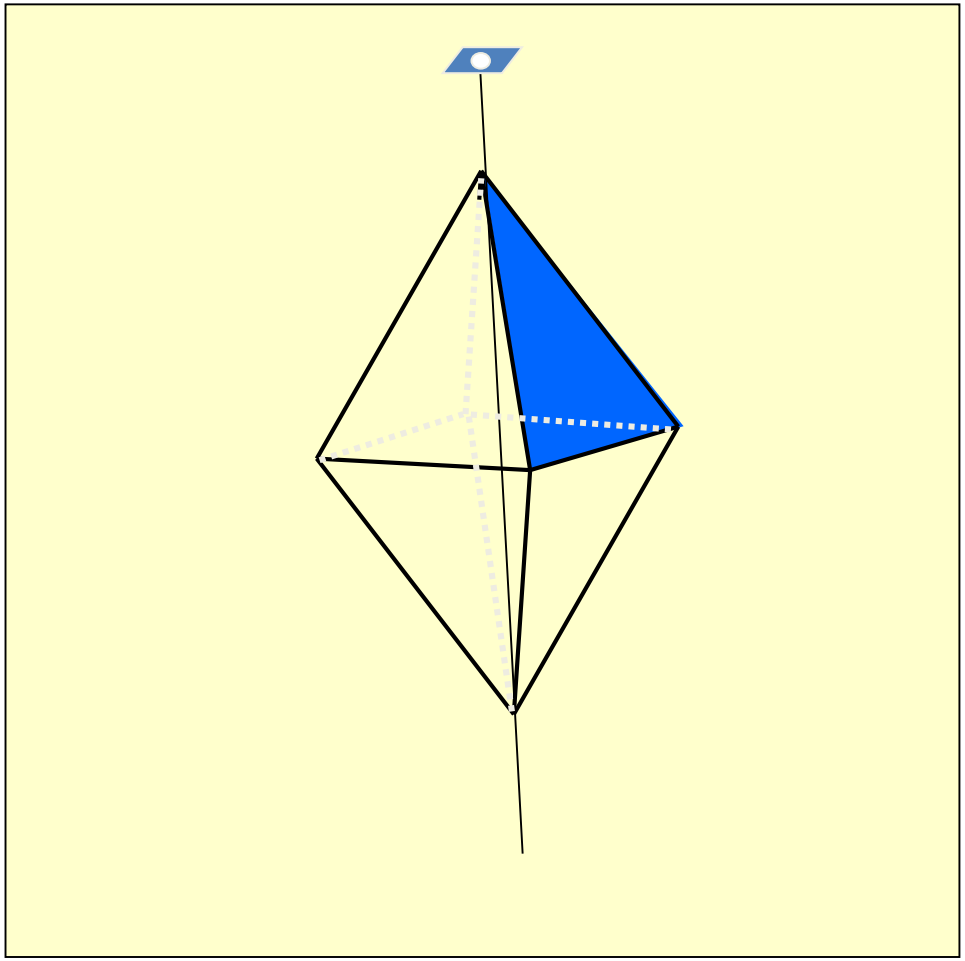


# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

d. 4-fold rotoinversion ( $\bar{4}$ )



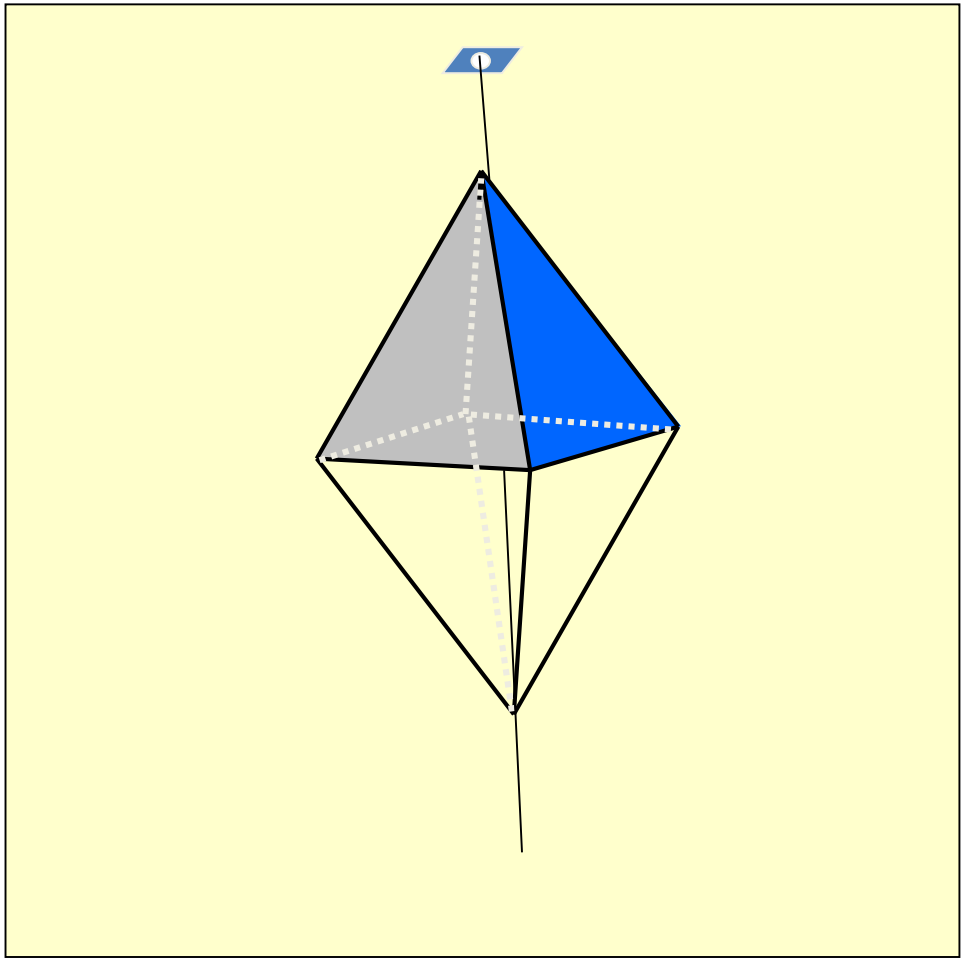
# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

d. 4-fold rotoinversion ( $\bar{4}$ )

1: Rotate 360/4



# 3-D Symmetry

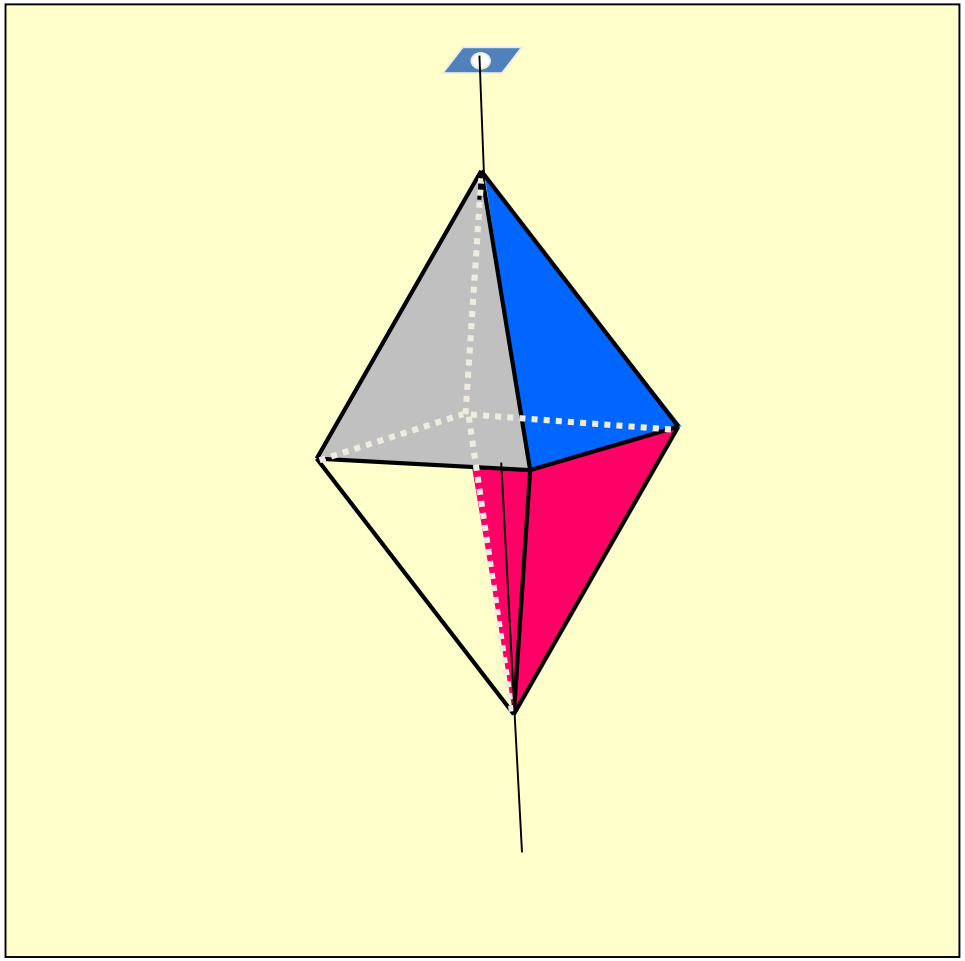
New Symmetry Elements

4. Rotoinversion

d. 4-fold rotoinversion ( $\bar{4}$ )

1: Rotate 360/4

2: Invert



# 3-D Symmetry

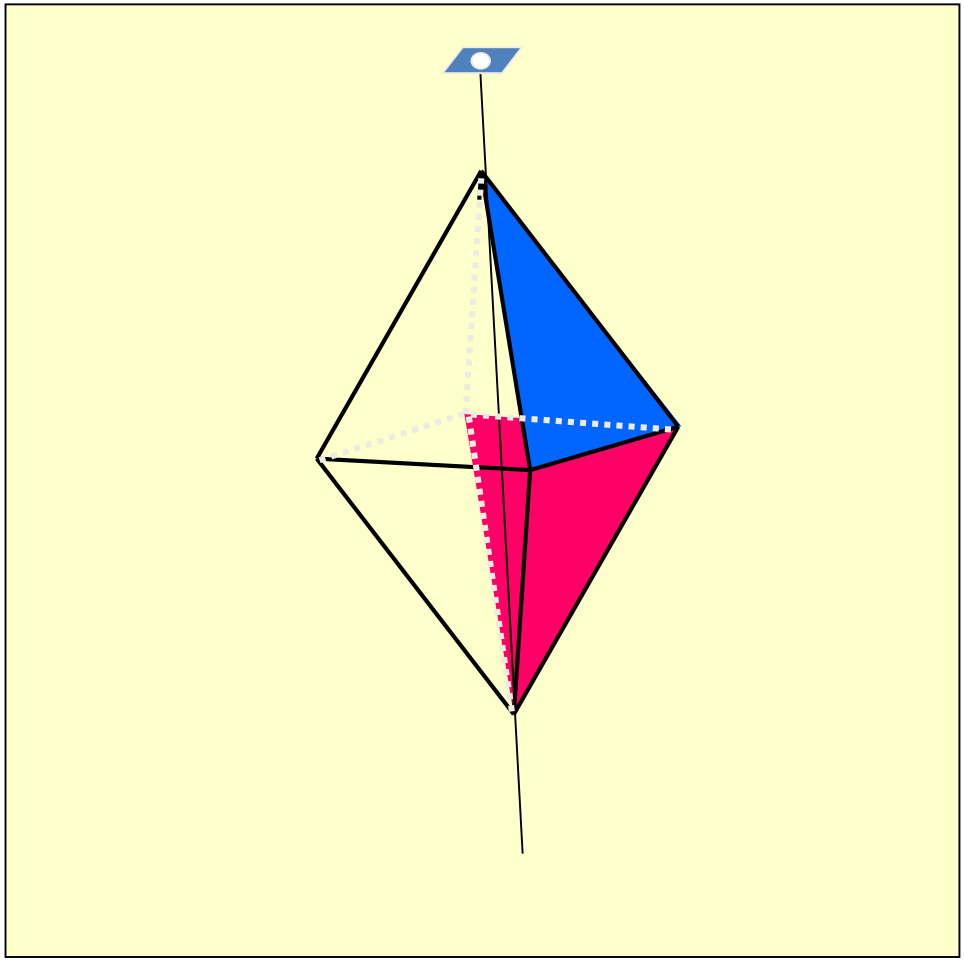
New Symmetry Elements

4. Rotoinversion

d. 4-fold rotoinversion ( $\bar{4}$ )

1: Rotate 360/4

2: Invert



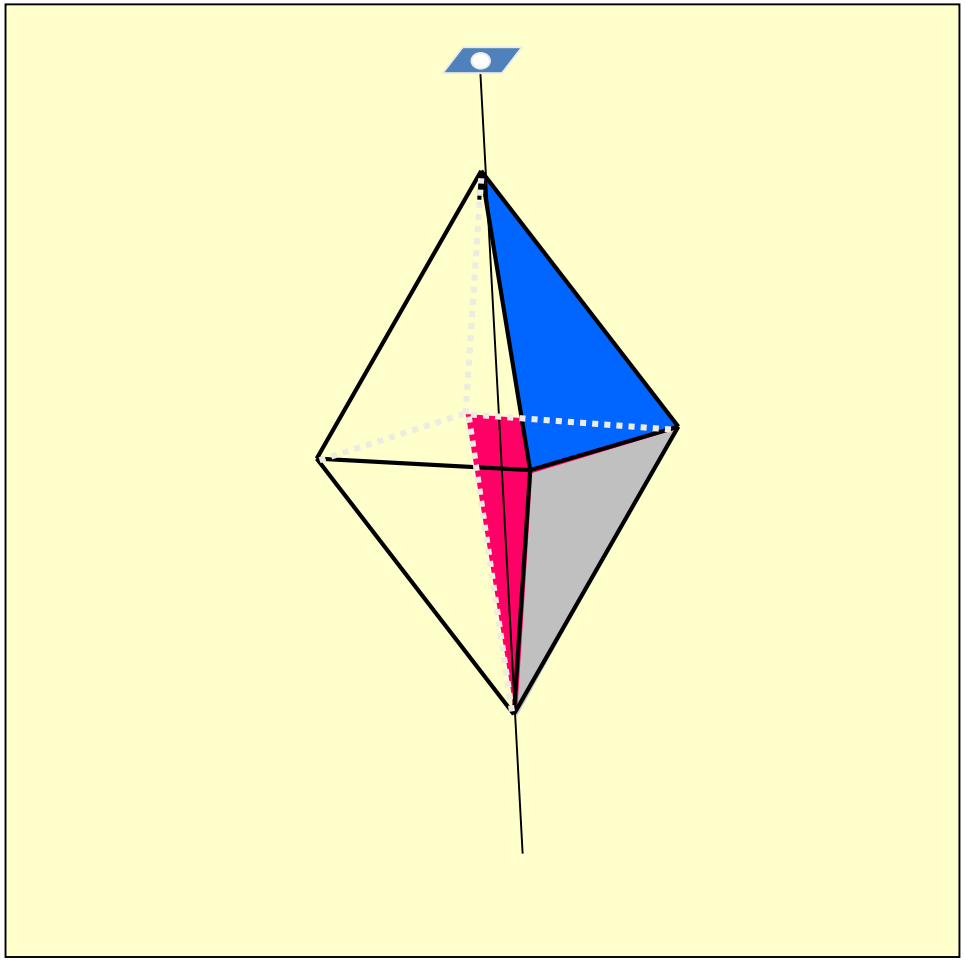
# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

d. 4-fold rotoinversion ( $\bar{4}$ )

3: Rotate 360/4



# 3-D Symmetry

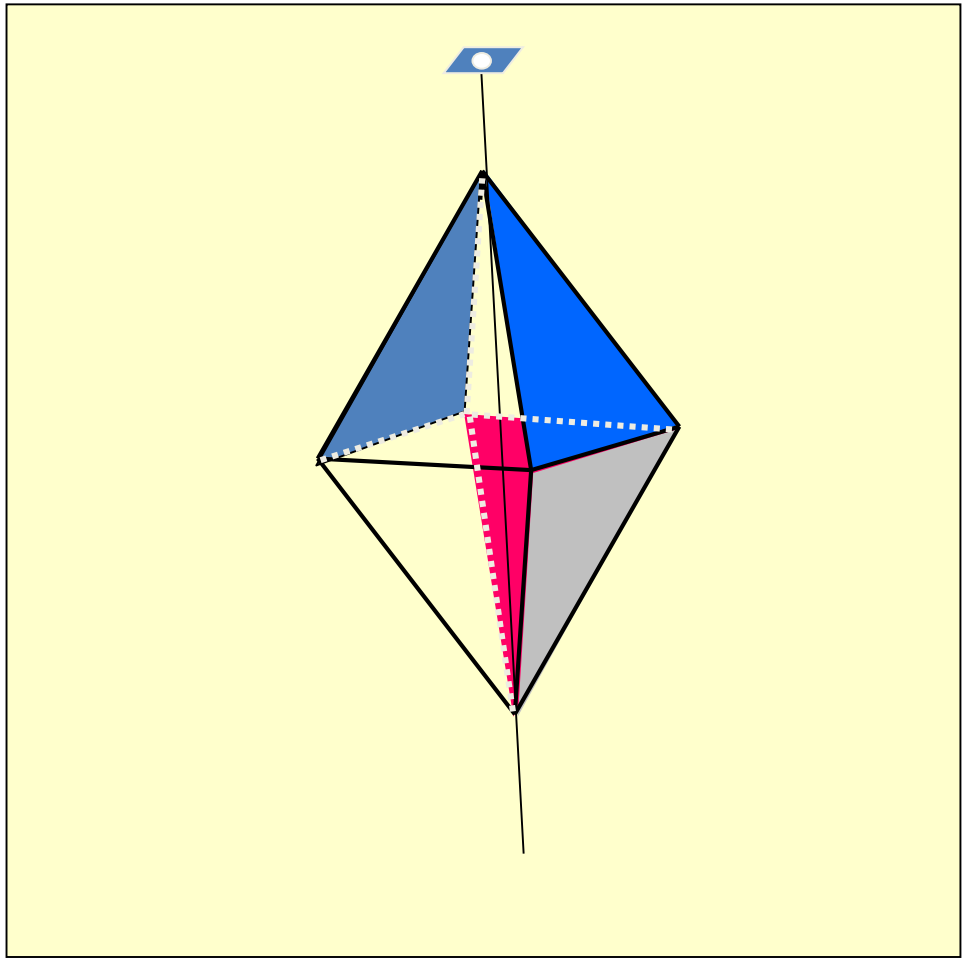
New Symmetry Elements

4. Rotoinversion

d. 4-fold rotoinversion ( $\bar{4}$ )

3: Rotate 360/4

4: Invert



# 3-D Symmetry

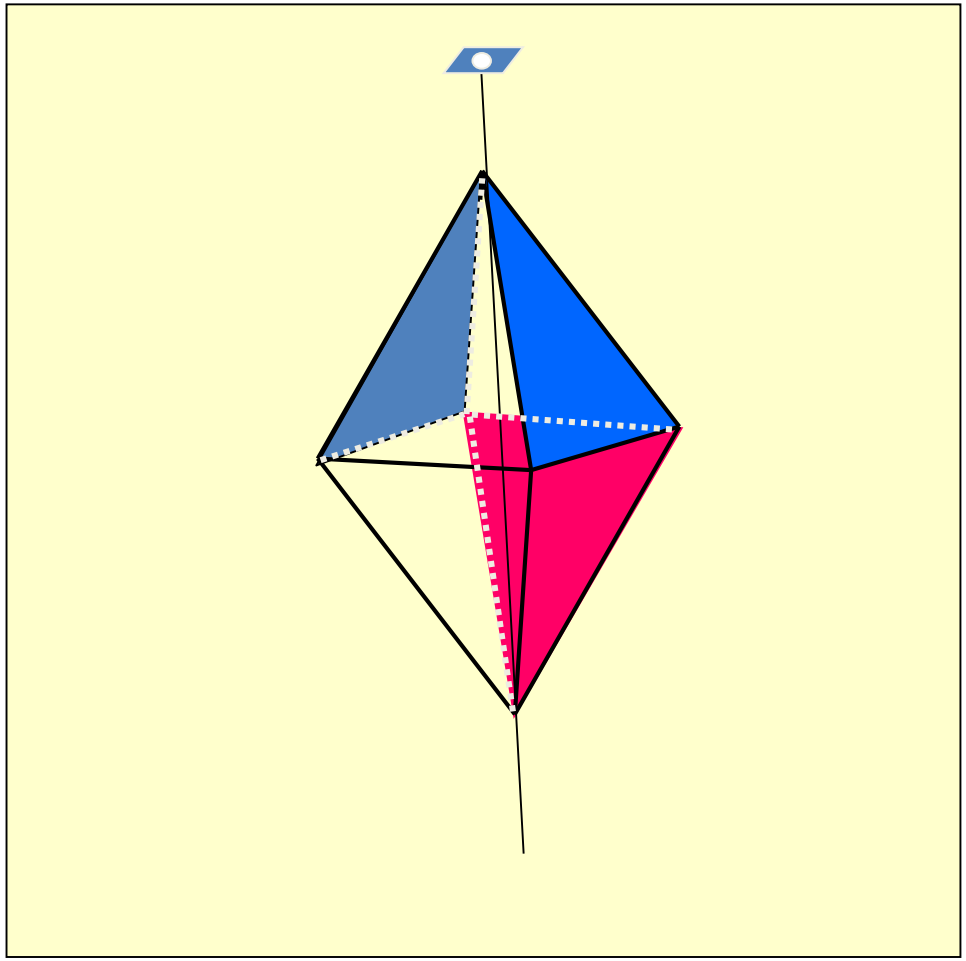
New Symmetry Elements

4. Rotoinversion

d. 4-fold rotoinversion ( $\bar{4}$ )

3: Rotate 360/4

4: Invert



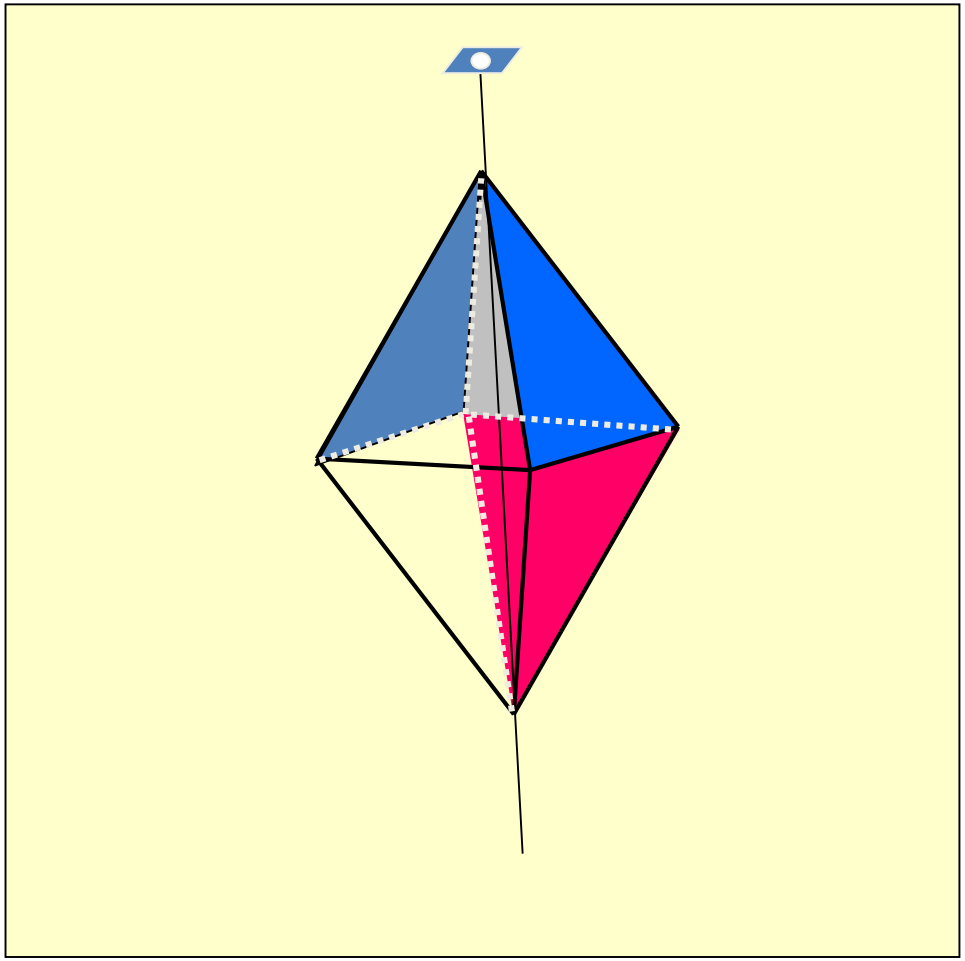
# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

d. 4-fold rotoinversion ( $\bar{4}$ )

5: Rotate 360/4



# 3-D Symmetry

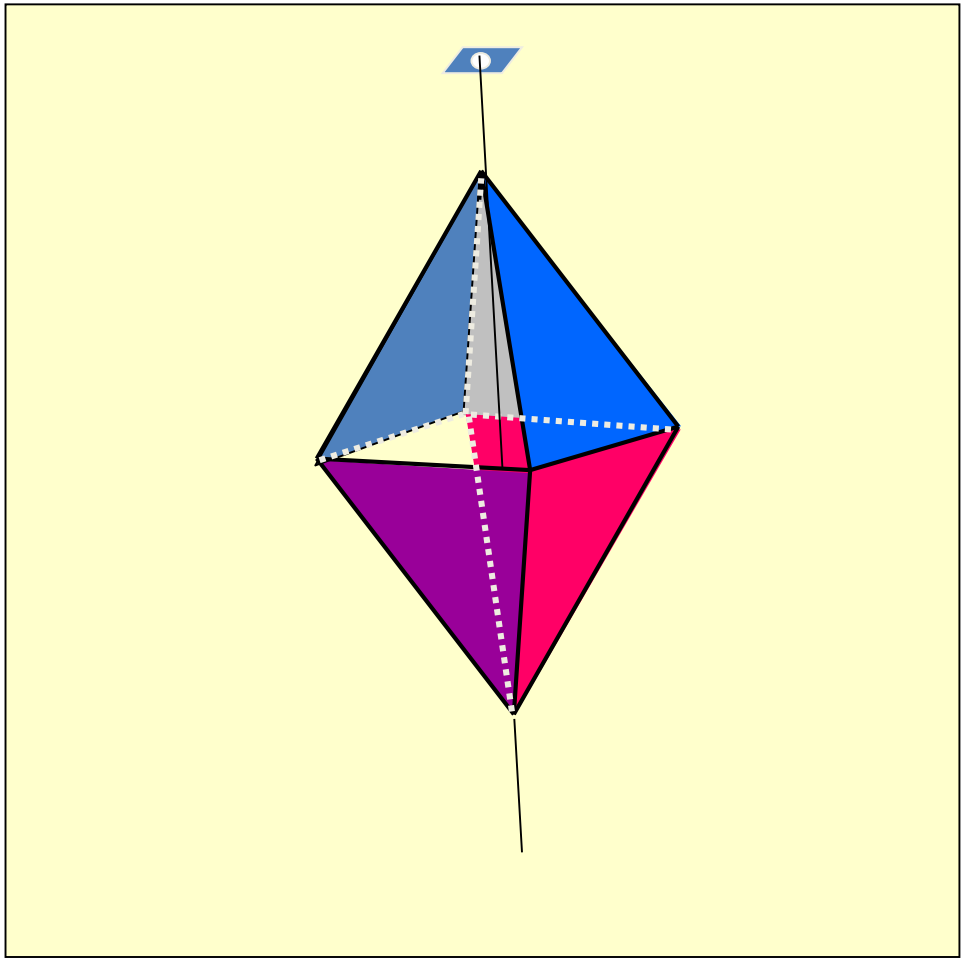
New Symmetry Elements

4. Rotoinversion

d. 4-fold rotoinversion ( $\bar{4}$ )

5: Rotate 360/4

6: Invert



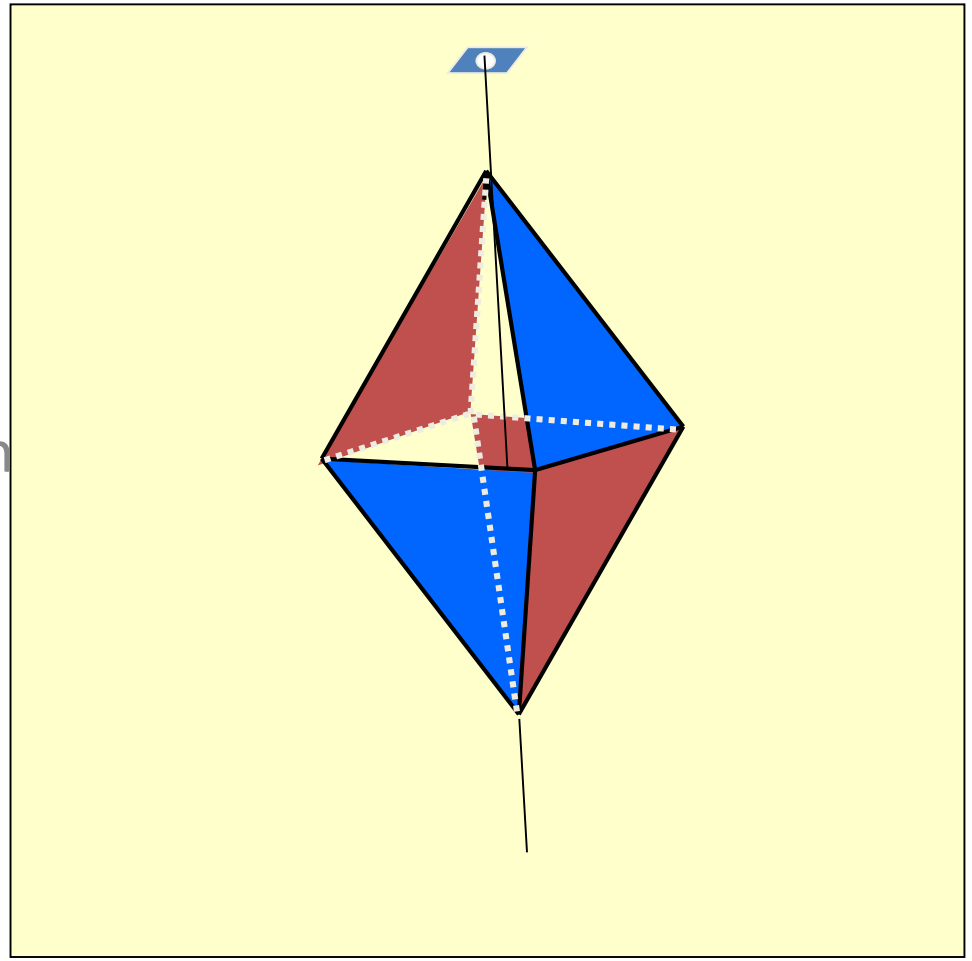
# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

d. 4-fold rotoinversion ( $\bar{4}$ )

This is also a unique operation



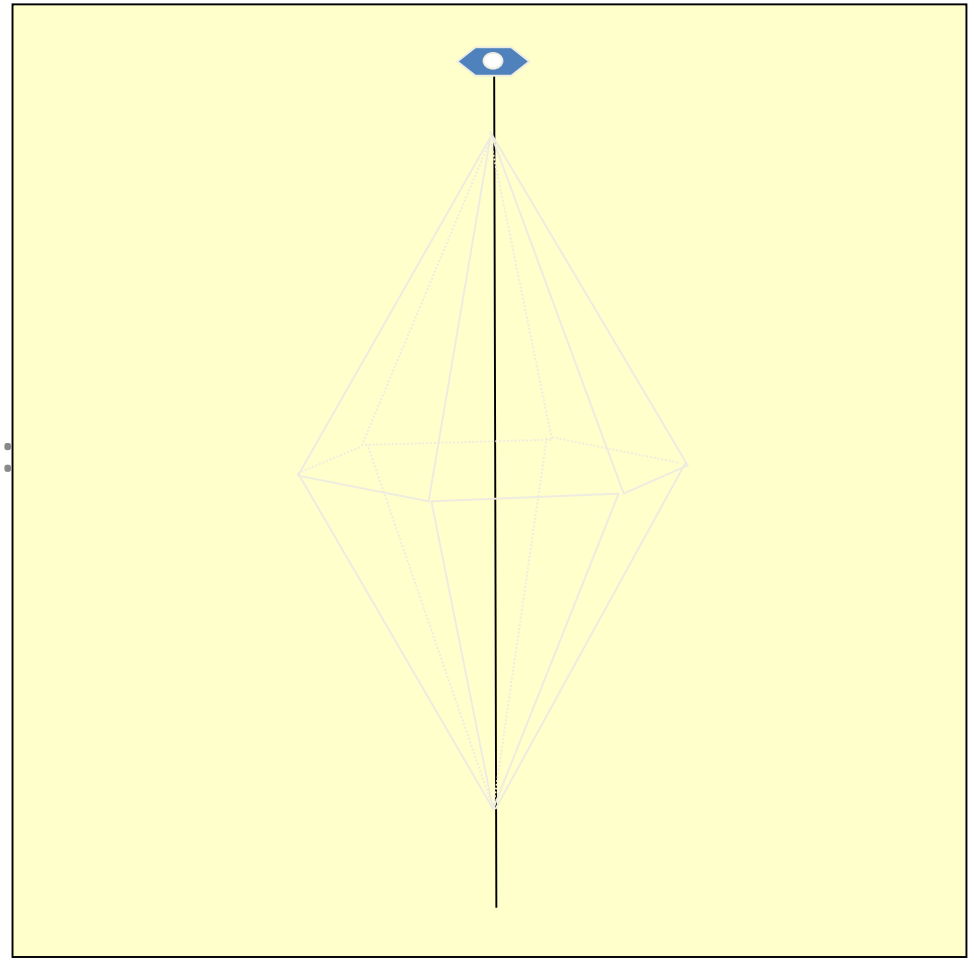
# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

e. 6-fold rotoinversion ( $\bar{6}$ )

Begin with this framework:

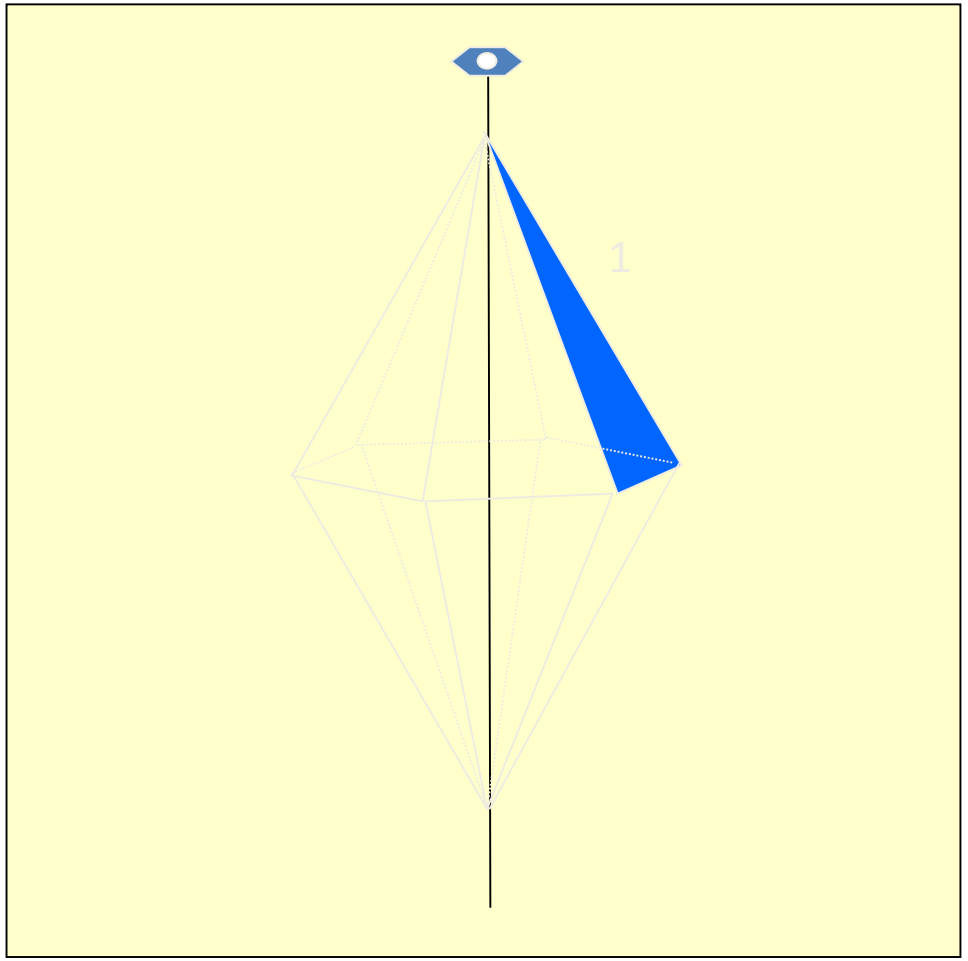


# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

e. 6-fold rotoinversion ( $\bar{6}$ )

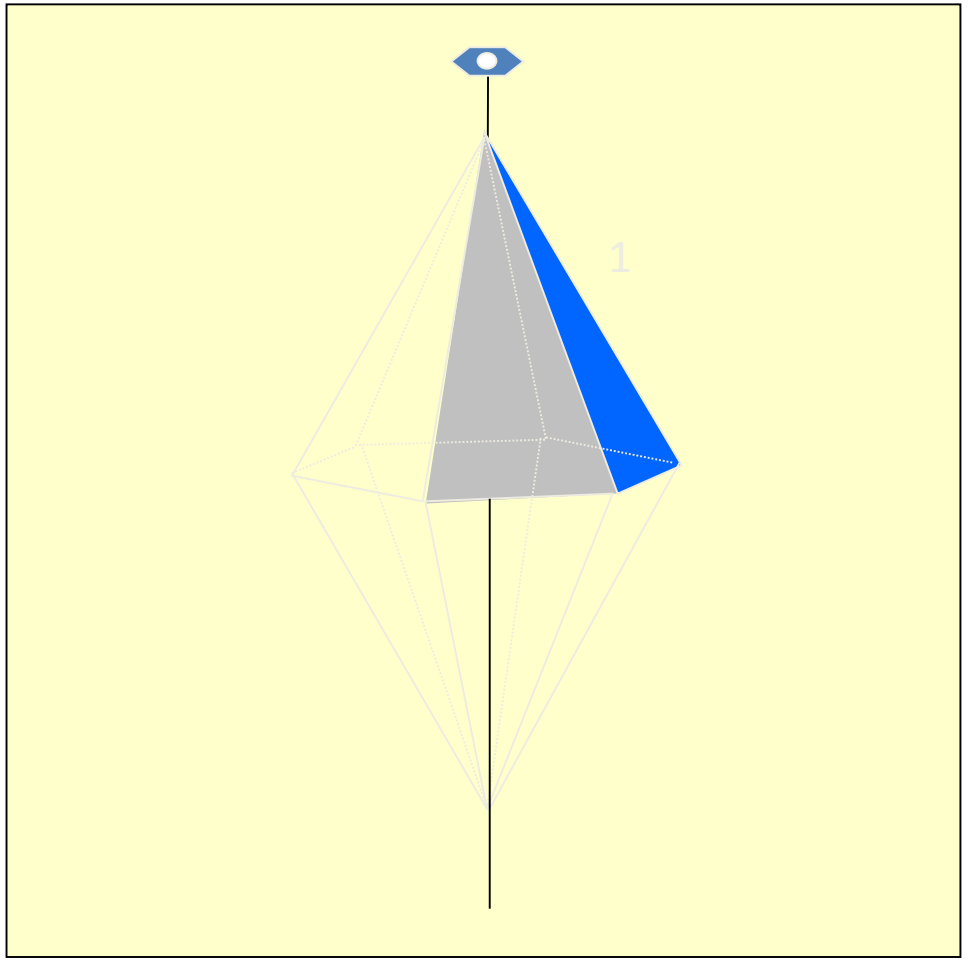


# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

e. 6-fold rotoinversion ( $\bar{6}$ )

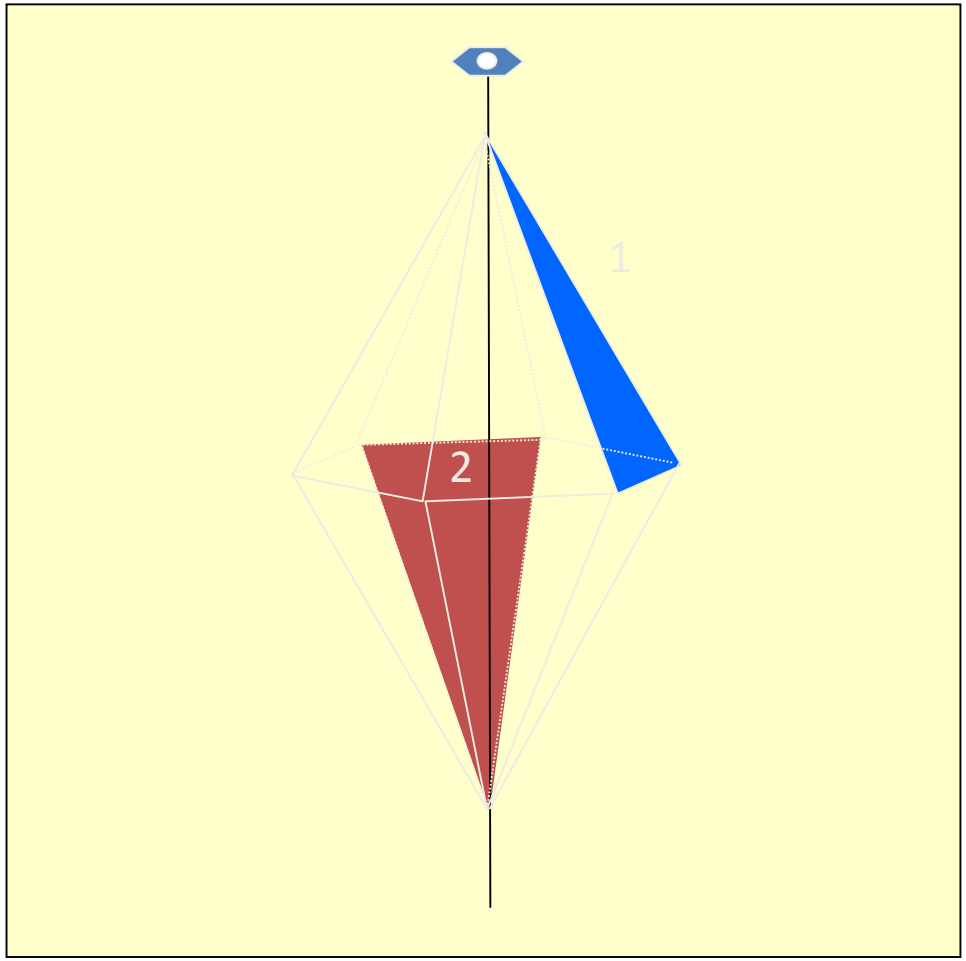


# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

e. 6-fold rotoinversion ( $\bar{6}$ )

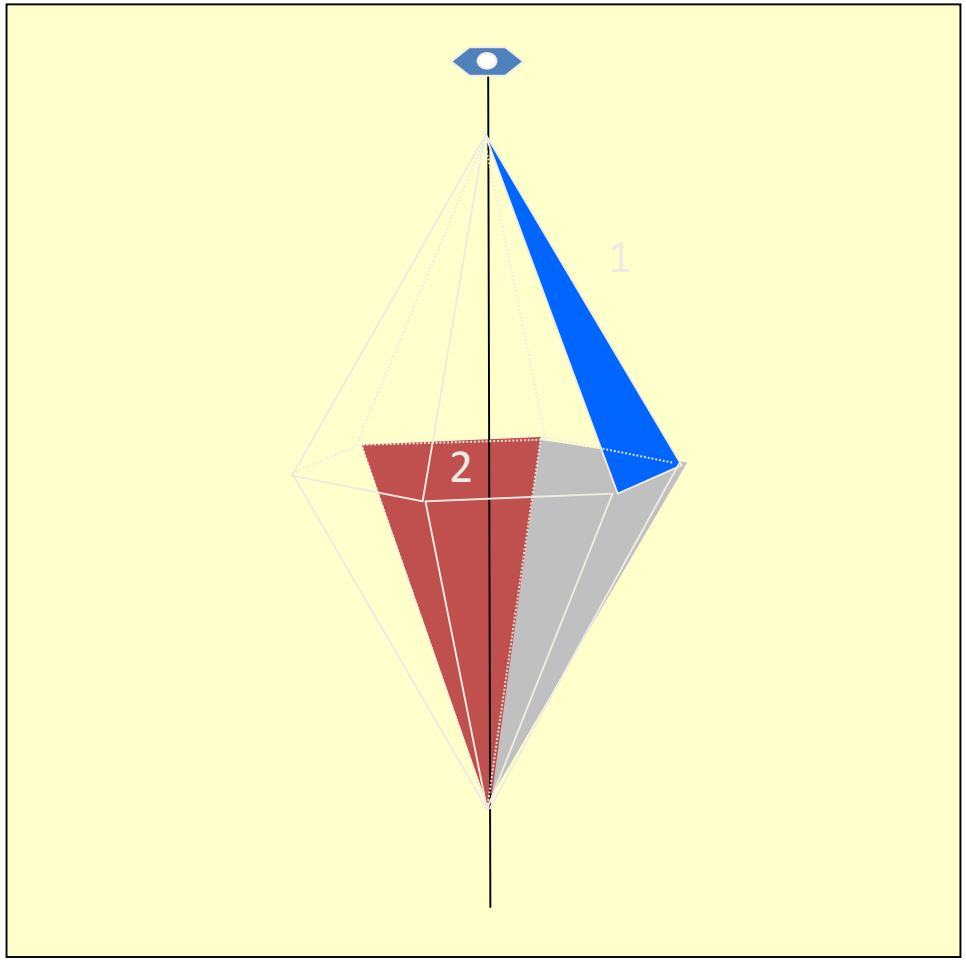


# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

e. 6-fold rotoinversion ( $\bar{6}$ )

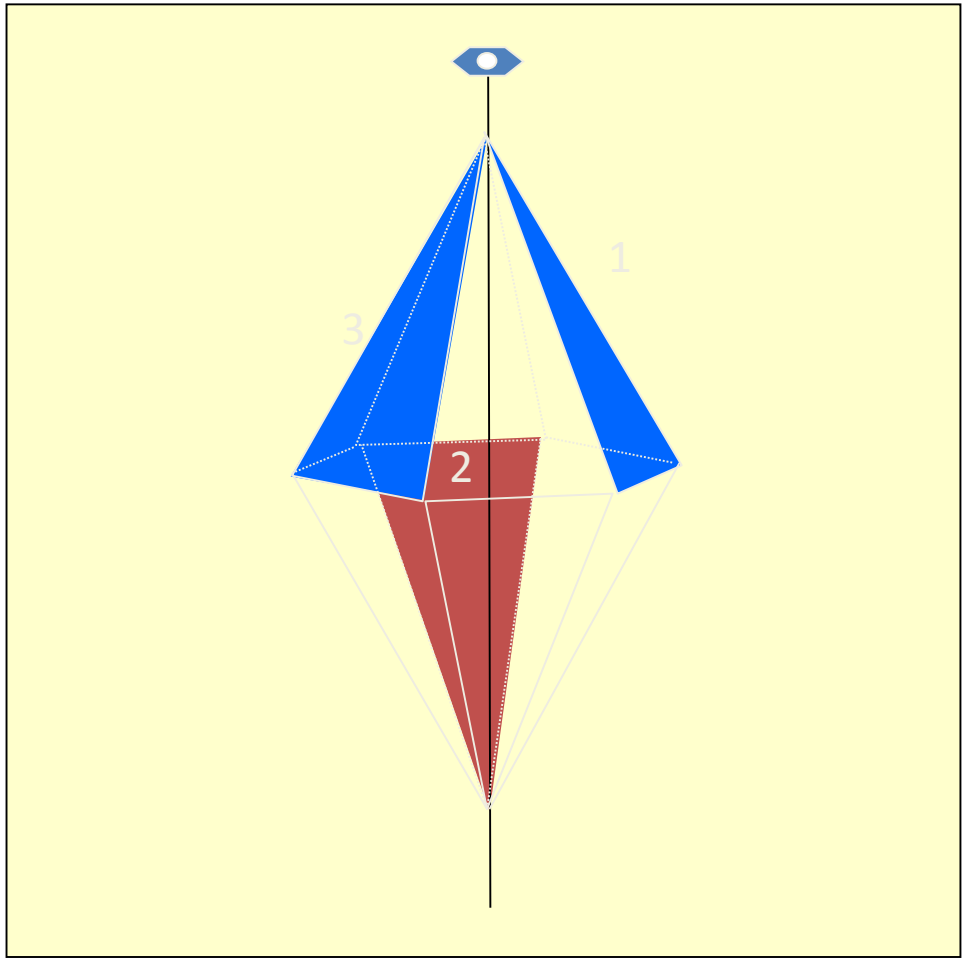


# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

e. 6-fold rotoinversion ( $\bar{6}$ )

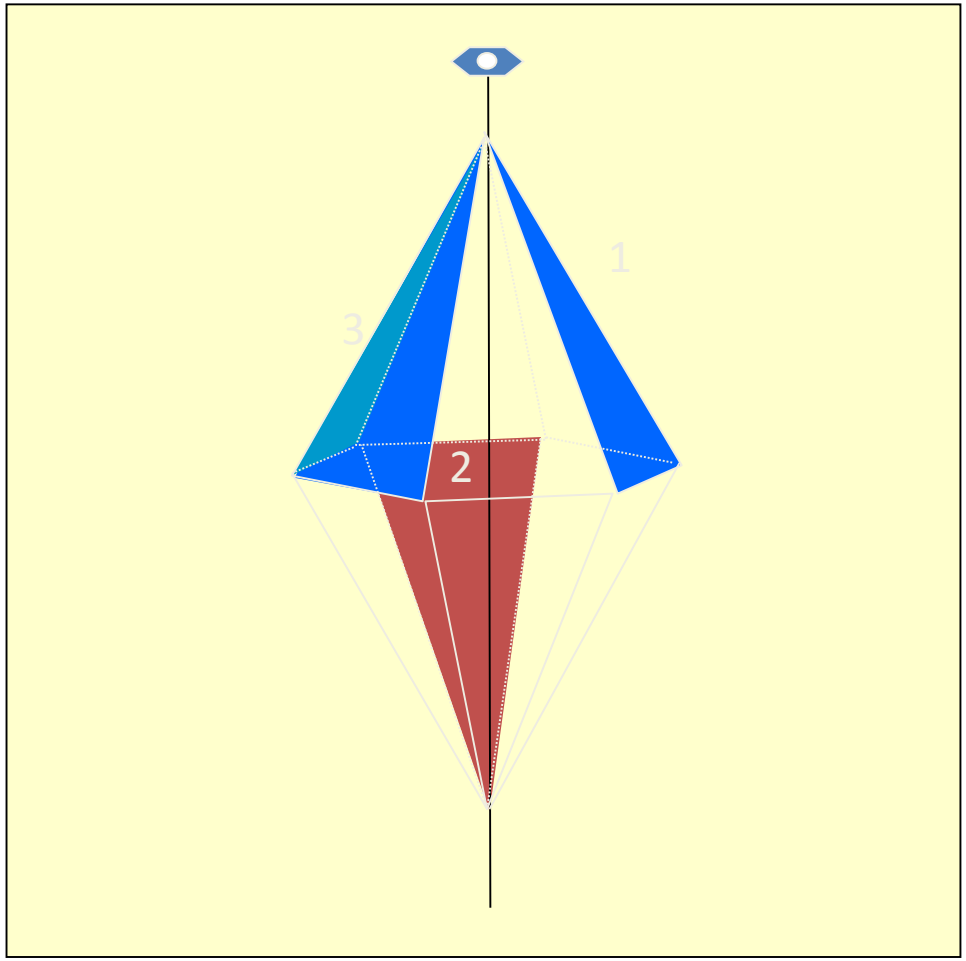


# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

e. 6-fold rotoinversion ( $\bar{6}$ )

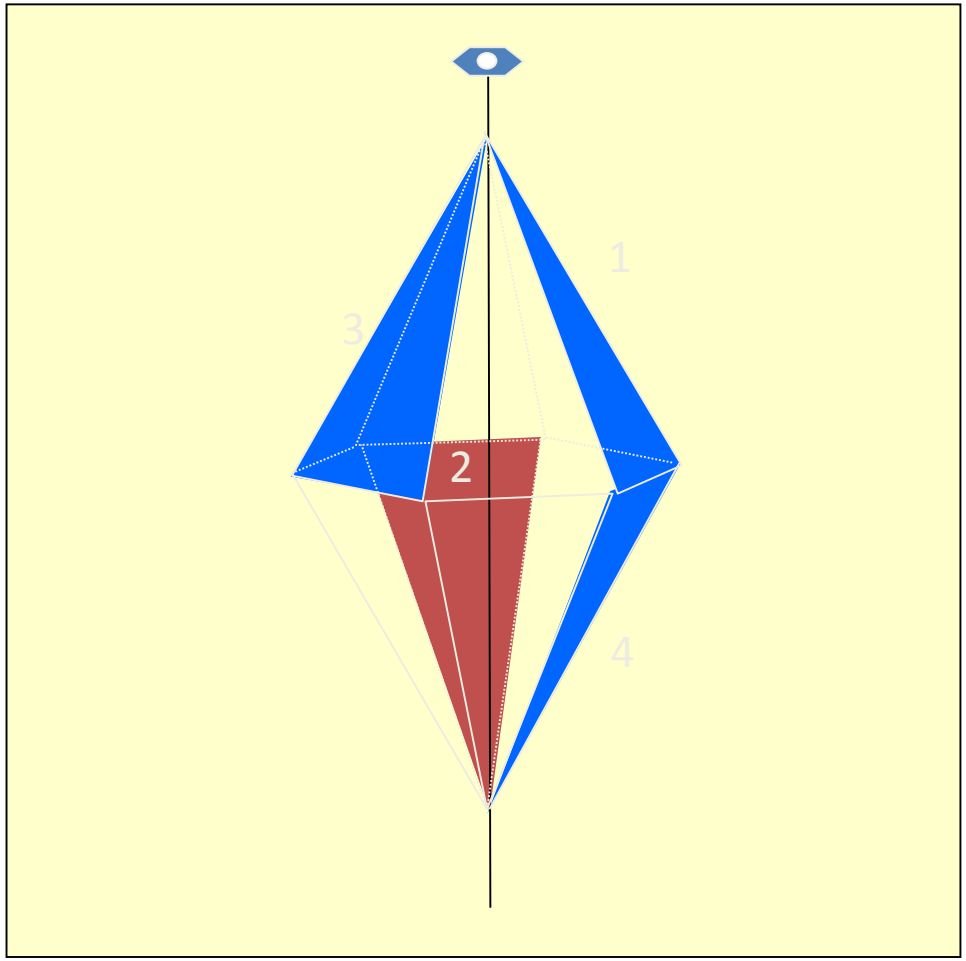


# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

e. 6-fold rotoinversion ( $\bar{6}$ )

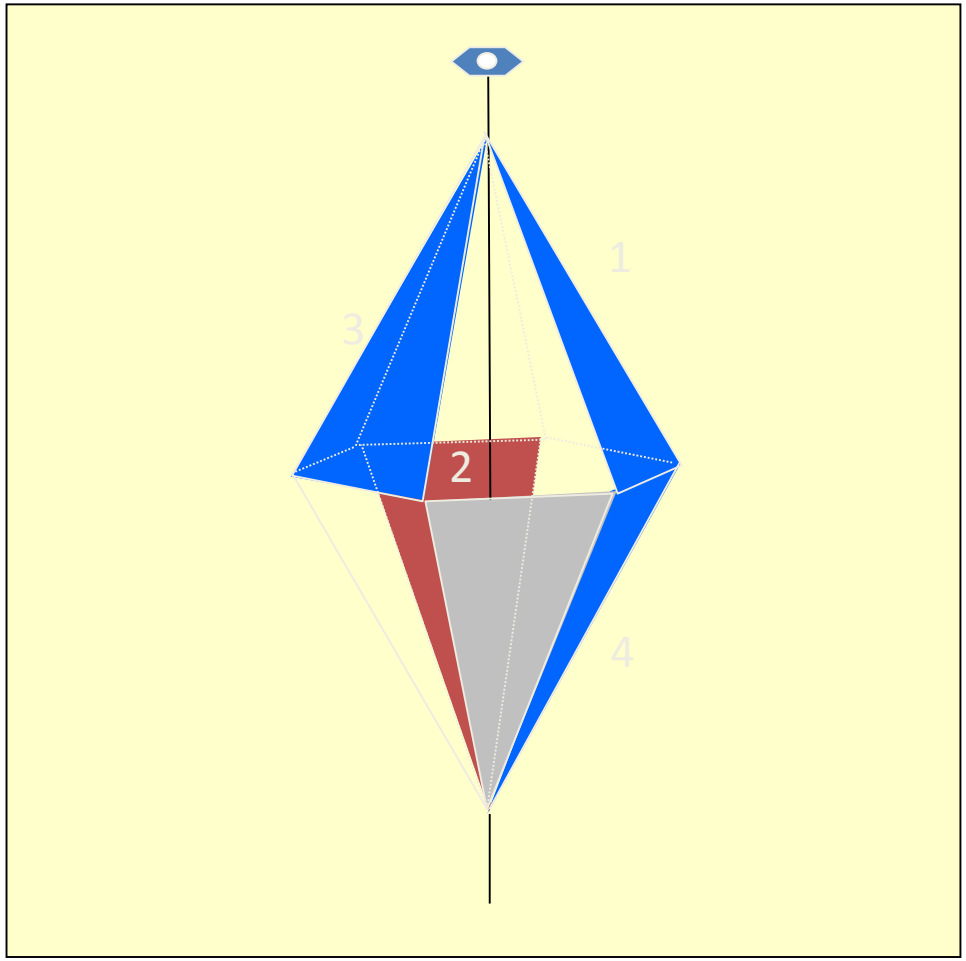


# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

e. 6-fold rotoinversion ( $\bar{6}$ )

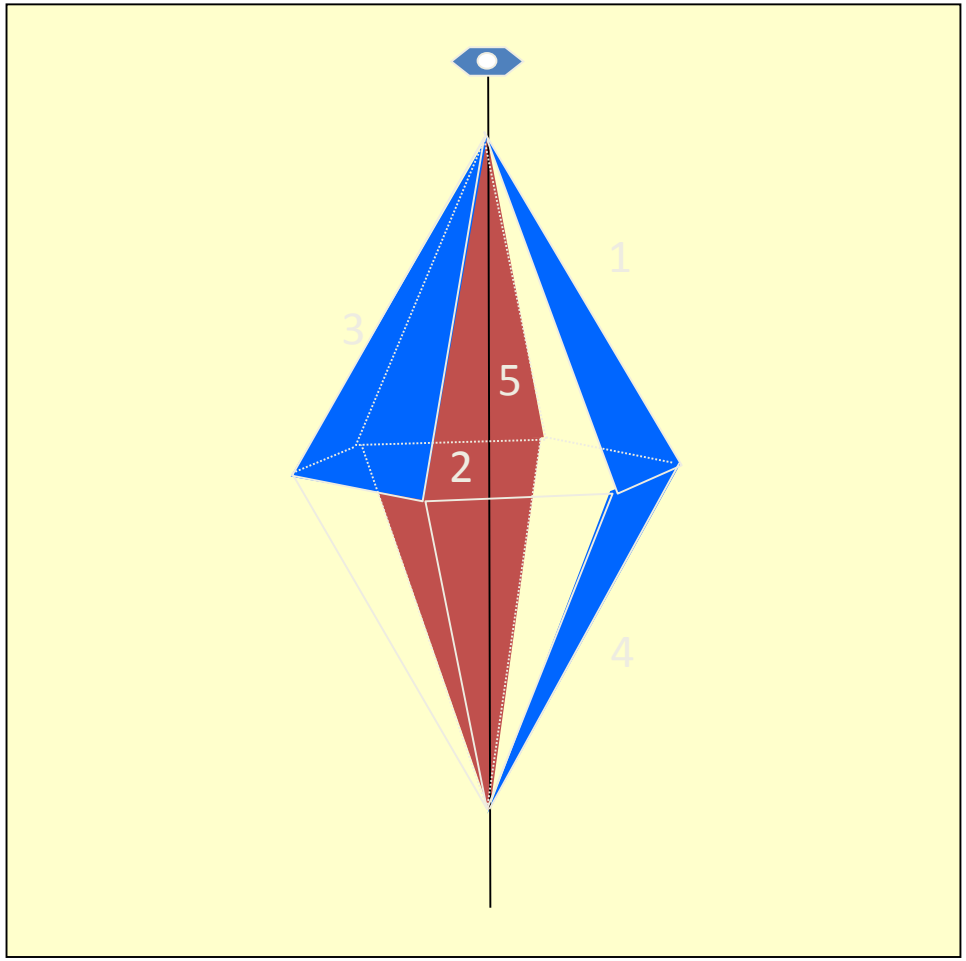


# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

e. 6-fold rotoinversion ( $\bar{6}$ )

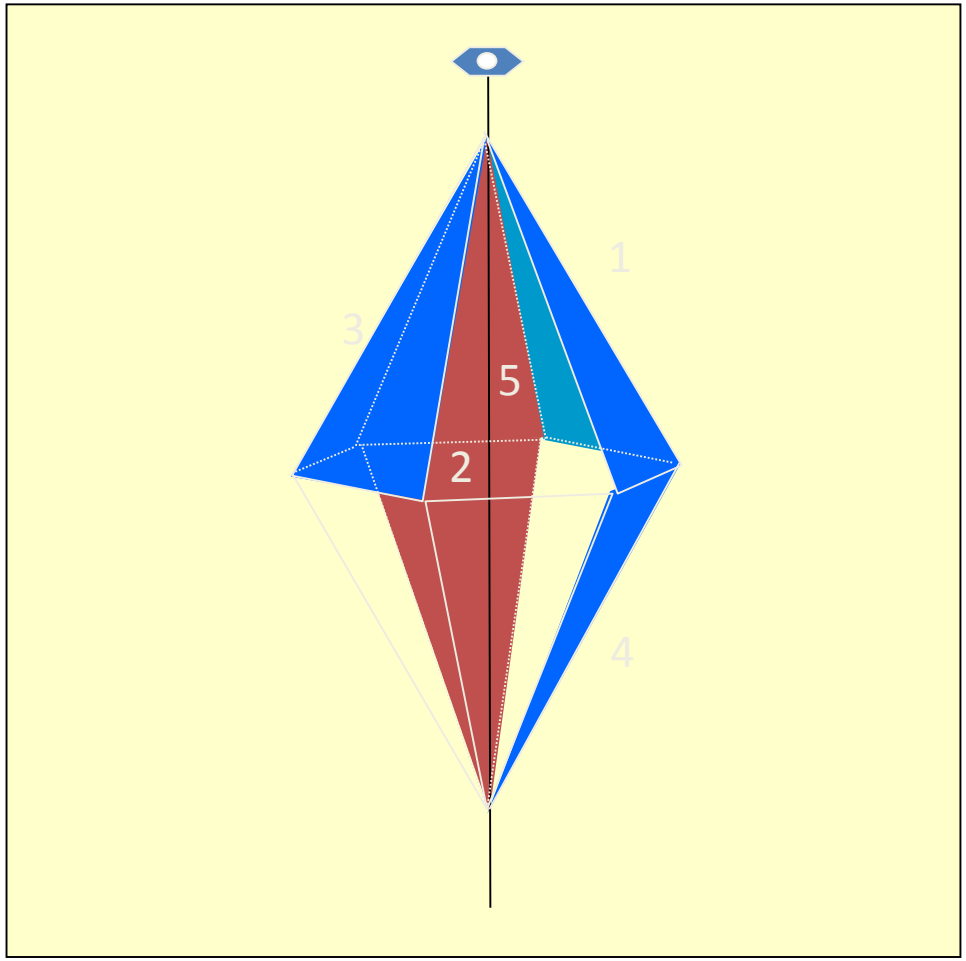


# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

e. 6-fold rotoinversion ( $\bar{6}$ )

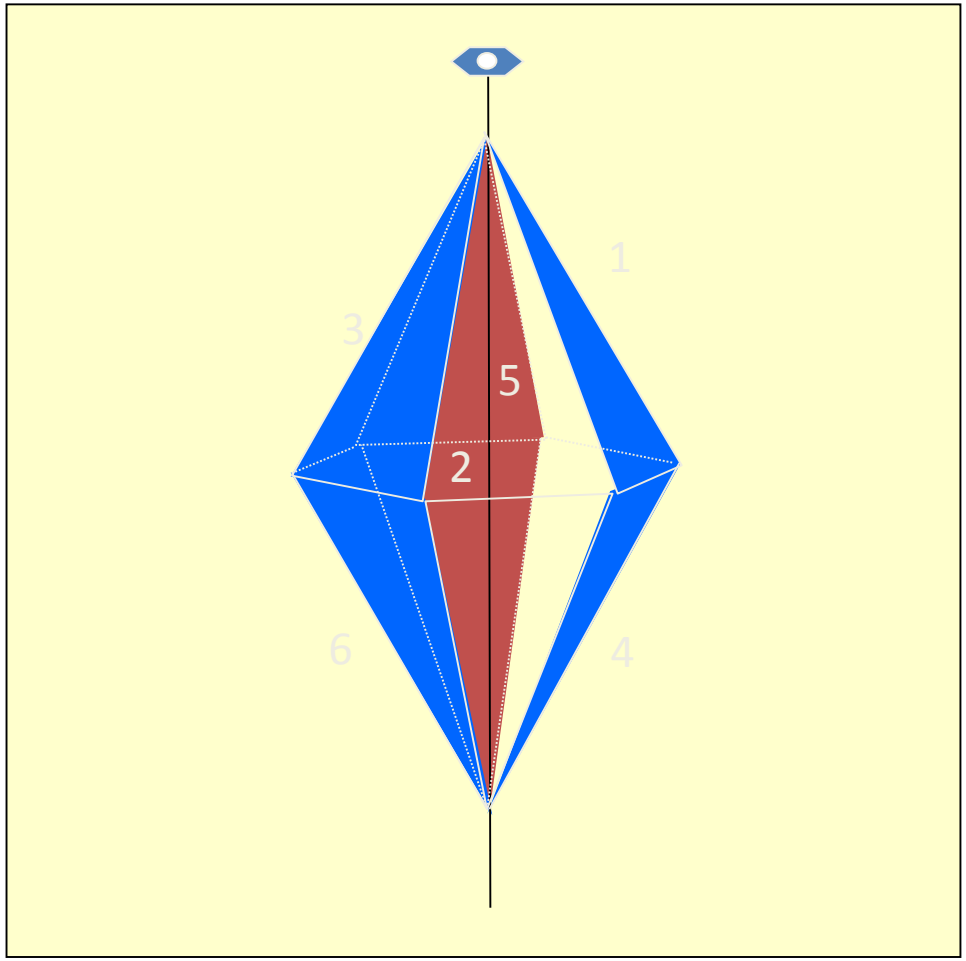


# 3-D Symmetry

New Symmetry Elements

4. Rotoinversion

e. 6-fold rotoinversion ( $\bar{6}$ )



# 3-D Symmetry

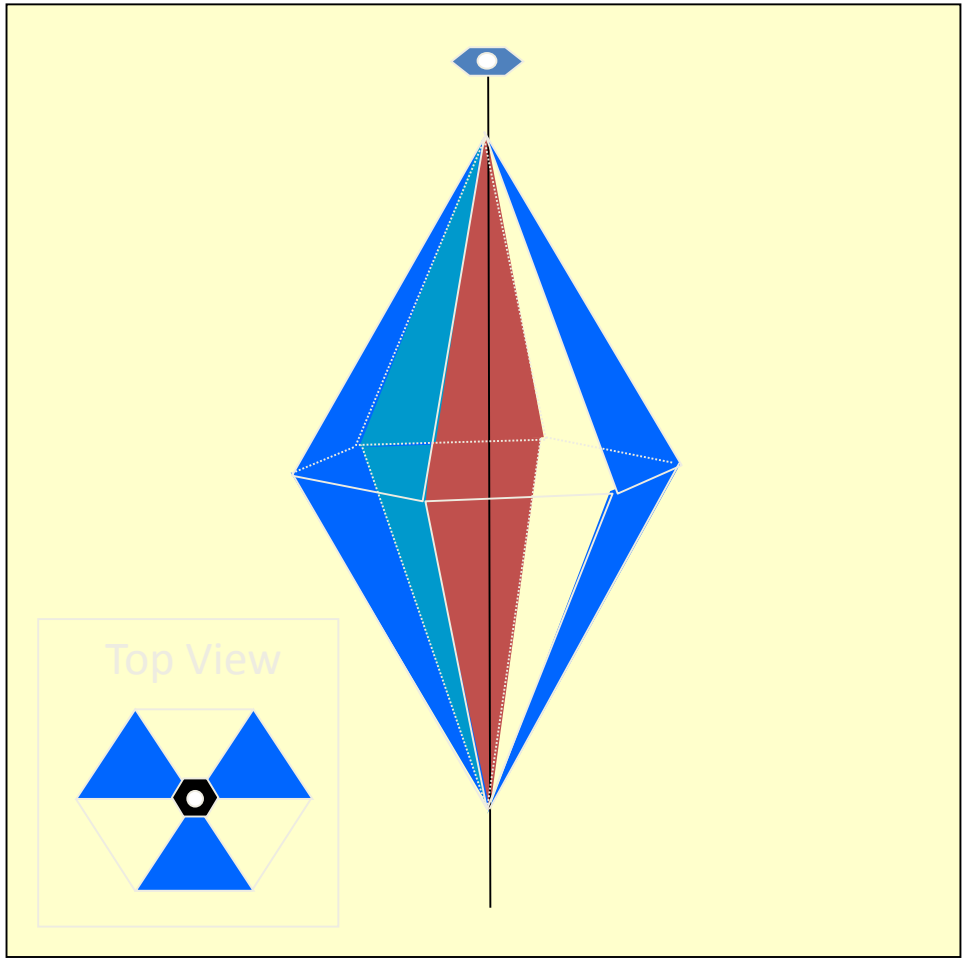
New Symmetry Elements

4. Rotoinversion

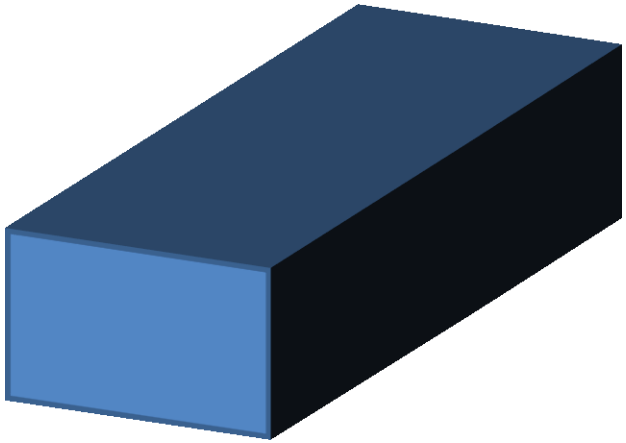
e. 6-fold rotoinversion ( $\bar{6}$ )

Note: this is the same as a 3-fold rotation axis perpendicular to a mirror plane

(combinations of elements follows)

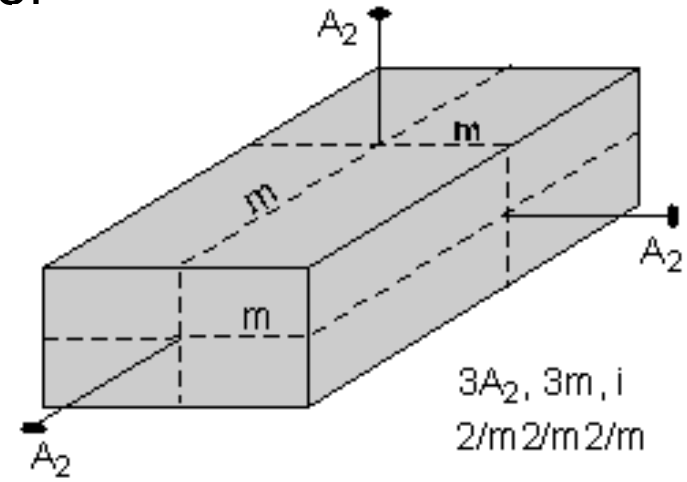


# Hermann-Mauguin Symbols



- To begin, write a number representing each of the unique rotation axes present.
- A **unique rotation axis** is one that exists by itself and is not produced by another symmetry operation.
- In this case, all three 2-fold axes are unique, because each is perpendicular to a different shaped face, so we write a 2 (for 2-fold) for each axis, A:

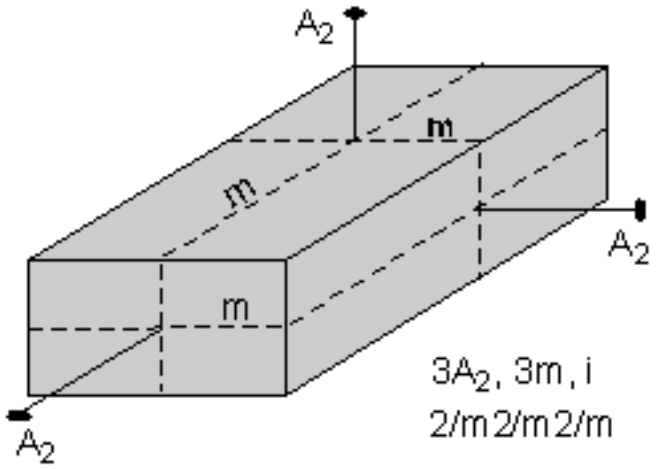
2 2 2



# Hermann-Mauguin Symbols

- Next we write an "m" for each unique mirror plane.
- Again, a **unique mirror plane** is one that is not produced by any other symmetry operation.
- In this example, we can tell that each mirror is unique because each one cuts a different looking face. So, we write:

2 m 2 m 2 m

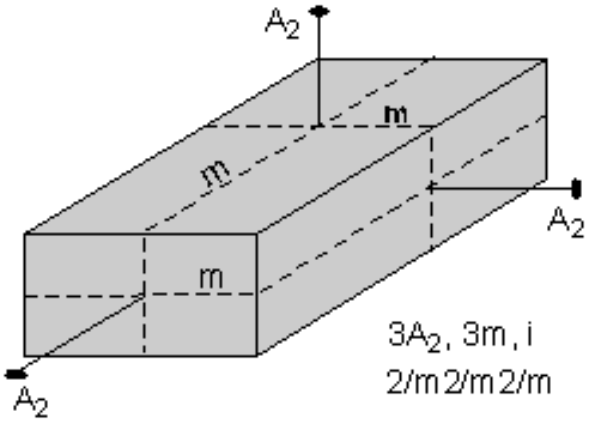


# Hermann-Mauguin Symbols

- If any of the axes are perpendicular to a mirror plane we put a slash (/) between the symbol for the axis and the symbol for the mirror plane. In this case, each of the 2-fold axes are perpendicular to mirror planes, so our symbol becomes:

  2/m 2/m 2/m

- The $D_{nh}$ Groups: $\frac{n}{m}\,\frac{2}{m}\,\frac{2}{m}$
- Briefly know as, $D_{2h}$ =*mmm*,
- $D_{4h}$ =4/*mmm*, $D_{6h}$ =6/*mmm*,



# 3-D Symmetry

3-D symmetry element combinations

a. Rotation axis parallel to a mirror

Same as 2-D

2 || m = 2mm

3 || m = 3m,  also  4mm,  6mm

b. Rotation axis ⊥ mirror

2 ⊥ m = 2/m

3 ⊥ m = 3/m,  also  4/m,  6/m

c. Most other rotations + m are impossible

2-fold axis at odd angle to mirror?

Some cases at 45$^{o}$ or 30$^{o}$ are possible, as we shall see

# 3-D Symmetry

3-D symmetry element combinations

d. Combinations of rotations

2 + 2 at 90° → 222 (third 2 required from combination)

4 + 2 at 90° → 422 ( “ “ “ )

6 + 2 at 90° → 622 ( “ “ “ )

# 3-D Symmetry

As in 2-D, the number of possible combinations is limited only by incompatibility and redundancy

There are only 22 possible unique 3-D combinations, when combined with the 10 original 3-D elements yields the 32 3-D Point Groups

# Lattice goes into itself through Symmetry without translation

| Operation | Element |
|---|---|
| Inversion | Point |
| Reflection | Plane |
| Rotation | Axis |
| Rotoinversion | Axes |

- The system of symmetry operations
- The point group symbol
- Or the Schonflies symbol (popular for group theory or spectroscopy)

# Symmetry Element Properties
(also true for 2D)

- Must satisfy conditions:
  - Two successive symmetry operations result in a further symmetry element: A * B = C
  - Associative rule: (A*B)*C = A*B*C
  - Identity element E: A*E = A
  - Every symmetry element possesses inverse: $A^{-1}$ *A = E
  - A*B = B*A
  - This leads to 32 distinct crystallographic groups
  - If you were also to allow translations, 7 --> 14, allowing 230 combinations, known as space groups